The Residual Causal Toolkit

Superposition as Mixed-Signal Geometry, a Residual Stream ISA, and Substrate-Agnostic Control Across Physical and Computational Buses

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8 September 2026

Companion repositories: *t2addonio/residual-causal-toolkit* and *t2addonio/residual-stream-grokking*

Abstract

We present a residual causal toolkit that treats residual streams as additive mixed-signal objects in which residual components exist in superposition with dominant carriers. The toolkit operationalizes a domain-agnostic protocol — mixture definition, contrastive extraction, a four-part causal filter (path-patch, zero, random, atom), and selective control — together with a Residual Stream ISA that names legal addresses on a bus that is never overwritten. Processors (transformer blocks, Fresnel propagators, Mueller matrices, frozen Fourier multipliers, lock-in banks) sit on that bus. They are not a new instruction set.

The same structural pattern transfers from discrete transformer residual streams through narrowband RF mixtures, pure classical optical residual fields (synthetic and real Mueller), nitrogen-vacancy color-center ODMR, Planck CMB residuals versus ΛCDM, multi-channel telemetry, EEG, audio STFT / CRAI isolation, an FNO-form continuum processor, and mid-circuit Pauli residuals of a hybrid quantum circuit. Across those domains residual structure is low-rank, often dominated by a pure second-moment component (energy^2 / E^2 / residual-power^2), and is selectively nullable while the carrier and the original stream remain intact.

Two honesty constraints are locked. Computer PASS is supporting evidence; on audio the product gate is a trained ear. A neural operator is a processor on the Residual Stream ISA, not a replacement ISA. A same-architecture Train A/B test on modular GPT-2 did not show that putting a residual READ in the training loop beats cross-entropy alone. Hardware now in hand includes an RTL-SDR Blog V4 (receive-only); the first owned-plant closed loop is defined and not yet run.

Keywords. residual streams · superposition · path-patch · mechanistic interpretability · optical Mueller calculus · NV-center ODMR · RF mixed-signal · Residual Stream ISA · CRAI

1. Introduction

In mechanistic interpretability the residual stream of a transformer is the central object in which features live in superposition: multiple features are encoded in overlapping linear directions of the same high-dimensional vector, and downstream circuits read out linear combinations of those directions [Elhage et al., 2021; Nanda et al., 2023]. Prior work isolated a causally necessary residual-stream component that persists after grokking and is nearly orthogonal to known algorithmic features [Taddonio, 2026a]. The present corpus asks a larger question: is that geometry an idiosyncrasy of neural activations, or is it the generic structure of an additive residual stream?

A narrowband RF observation is a linear mixture of carriers, interferers, and residual products including IM3. A coherent optical field is a linear superposition of a main beam and residual spatial, modal, or polarization components. An ODMR spectrum of an NV center is a carrier resonance plus residual shifts induced by strain, magnetic field, or temperature. Multi-channel telemetry is a dominant process plus residual mechanical or performance-decay components. A Planck power spectrum is a best-fit ΛCDM carrier plus a residual. In each case a residual stream can be written

$$x = carrier + residual\ (+\ noise)$$

and the residual component remains causally necessary for residual-dependent observables even after the carrier is accounted for. This paper unifies four earlier working documents — the optical residual-streams paper, the superposition synthesis, the quantum residual-intervention note, and the Residual Stream ISA one-pager — with the campaign that followed through 8 September 2026: 14-D pure-quadratic lifts, parallel residual ports, a logical residual layer, GPT-2 Phases A–D (online write-hook, 16/16 seeds), CRAI audio isolation, an FNO-form continuum processor, a failed trainer A/B, and first IQ from an owned RTL-SDR V4.

The contribution is methodological and experimental. We do not claim that every residual-like signal in the wild is path-patchable. We claim that when a residual component is present in an additive mixture and the ensembles are properly defined, the same addressing filter recovers and controls it, and that the original stream can be left untouched while parallel ports do the work.

2. The Residual Causal Toolkit

2.1 Core principle

A residual stream is treated as an additive mixture of a dominant carrier plus one or more lower-dimensional residual components. The residual is nearly orthogonal, or selectively isolatable, from the carrier and remains causally necessary for residual-dependent observables. Two complementary recovery routes are used: (i) contrastive extraction of a residual direction or subspace from mean(signal) − mean(held-out), and (ii) sparse-dictionary / SAE-style recovery that places the residual direction as an explicit dictionary atom whose coefficient can be ablated.

2.2 Protocol steps

Step 1 — Define the mixture. Name the residual stream. Construct a signal ensemble that contains the residual and a held-out ensemble that lacks it but shares carrier statistics. Name the residual-dependent READ (accuracy, residual power, local intensity, strain proxy E, compressor-decay power, isolation SI-SDR, answer log-probability).

Step 1b — If and only if the residual is a field $u(x)$ on a domain Ω. Keep Ω (metric, boundary, causal order) separate from the fiber F at each point (amplitude, Stokes, residual channels). Name the processor G if any. Open a kernel port $\Delta R = R_{sig} - R_{held}$ only when a multiplier exists. Do not flatten Ω to feed an FFT if the room had walls or lookahead was illegal. Skip 1b on vector buses.

Step 2 — Contrastive extraction. Mean difference, single-feature probe arms, SVD as a diagnostic, a 14-D pure-quadratic lift (x, x^2, $x_i x_j$) with no bias and no centering, residual-seeded dictionary / SAE. Do not treat a co-occurrence SVD as a 2-D PHASE core when residuals always appear together; that matrix is rank-1.

Step 3 — Causal filter. Keep a port only if all four fire: path-patch of the coefficient toward the held ensemble moves the READ toward held; zero-ablation collapses residual power; a matched random direction or subspace leaves residual metrics intact; residual-atom ablation is selective against random-atom ablation. Complement and original stream S are never written except at an explicit write-hook site, which is checksummed.

Step 4 — Selective control. STORE / PHASE / BEAM on a copy, or an online write-hook of the form $r \leftarrow r - c\,d + c_{target}\,d$ at a chosen site (last token, source plane, residual coefficient).

Step 5 — Validate. Residual-dependent READ moves; random intact; S checksum unchanged except at the controlled write.

2.3 Path-patch as mediation test

For a residual direction d and stream (or field) E, the residual coefficient is c = ■d, E■. Path-patch replaces c with a new coefficient c′ (held-out mean or zero):

$$E' = E - c\,d + c'\,d$$

leaving the carrier intact. If residual-dependent observables move to the held-out regime under residual-coefficient path-patch, and remain near baseline under random-direction path-patch, the residual component is causally mediating the residual-dependent difference between ensembles. This is the primary causal claim of the toolkit.

2.4 Residual Stream ISA

The residual stream is not a computer. It is RAM: a bus that the environment already writes. Layers, mixers, Hamiltonians, and optical paths are the processors. The ISA names addresses on that bus, proves they are real, and issues one instruction that rotates two of them without touching the complement. It is not a QPU (PHASE is one qubit / two modes), not a CPU/GPU/TPU, and not a computer built from analog energy. Energy is a power rail. A clock is one address.

Op	Action	Must hold
LOAD	Sample S from the source	Dimensional depth ≥ 2
EXTRACT	Contrastive + SVD → (u, v)	Two ensembles exist
FILTER	Path / zero / random / atom	Random must fail
STORE c	resid ← resid – c u + c* u	Complement untouched
PHASE ϕ	Givens rotate of (α, β) in span{u,v}	■r■ and plane power flat
BEAM θ	SU(2) mix of the two arms	Same unitarity
READ	Residual-dependent observable	Score the task, not the pretty direction
FAULT	Reject the port	Write moved the complement, or random also worked

Table 1. Residual Stream ISA. Two legal addresses. One rotate. A fault if the address was never legal.

Eligibility fails if the source is a single scalar, if there is no contrast (only energy), if the residual is rank-1 with no plane after FILTER, if measurement is destructive-only with no hook, or if the timescale does not match the task.

2.5 Methodological locks

- Path-patch is the core causality test. Correlation is not enough.
- Original stream S is RAM, not a scratch pad.
- Computer PASS is supporting evidence. On audio the product gate is a trained-ear isolation bar.
- Toolkit DSP first. No neural net, no weights, no GPU inference in CRAI.
- 96 kHz live audio: STFT insert latency is illegal. Causal lock-in only.
- Neural operators process the bus. They do not replace FILTER.
- When residuals co-occur, use single-feature probe arms. Co-occurrence SVD-1 is not a 2-D PHASE core.
- Hard-zero of a template that still contains carrier punches a hole; G will diffract that hole back into the residual READ.

3. Path-Patch Mediation Across Domains

Path-patch of residual coefficients moves residual-dependent observables toward the held-out regime in every domain tested. Zero-patch collapses residual power to numerical zero. Matched random-direction or random-subspace patches leave residual metrics essentially intact. Carrier coordinates remain relatively protected.

Domain / experiment	Orig residual	Path → held	Zero	Random
Optical P1 single-mode	53.31	15.95 (held 16.11)	~0	53.29 intact
Optical P3 Jones	13.72	0.15 (held 0.13)	~0	intact; H-pol protected
RF-P1 single residual	320566	5.95	~0	320544 intact
RF-P2 subspace	612322	340	~0	611386 intact
NV single E strain	E = 94.8 MHz	3.28 MHz (held)	0	intact
NV multi-NV subspace	0.315	0.0075 (held)	0	0.302 intact
Telemetry real naval	1.374	0.0084	~0	1.295 intact
Audio residual tone	1018.09	0.100 (held 0.100)	0	1010.93 intact
EEG question residual	66.3	1.35	0	intact
GPT-2 Phase D Δacc	baseline	path −22.0%	zero −16.1%	random −0.2%
Logical NV consensus	829.86	3.19	0	filter PASS 20/20
Continuum P1 field port	115.30	7.36	~0	115.29 intact

Table 2. Cross-domain residual causal evidence. Path-patch is the mediation test; zero is the necessity test; random is the selectivity test.

4. Pure Classical Optical Residual Fields

The first physical transfer after the transformer / RF work was a pure classical optical residual field. Color centers were deferred on purpose until the classical case was fully exercised. Synthetic ensembles used complex scalar fields on a 64×64 grid (or Jones / Stokes equivalents). Carriers were Gaussian main beams. Residuals were offset spatial modes, multi-mode subspaces, vertical Jones residuals on horizontal carriers, residual Stokes components, or residual Mueller transformations (weak diattenuator or linear retarder).

4.1 Static single-mode and Fresnel propagation

Contrastive residual directions were recovered for static single-mode residuals. Projection ablation ($\alpha = 1$) nulls residual power to numerical zero; random directions leave residual power intact; $\alpha = 2$ sign-flips the component. After free-space Fresnel / angular-spectrum propagation ($z = 180$, $\lambda = 2$) the residual remains contrastively recoverable and selectively nullable. The residual diffracts under unitary evolution but its causal structure survives.

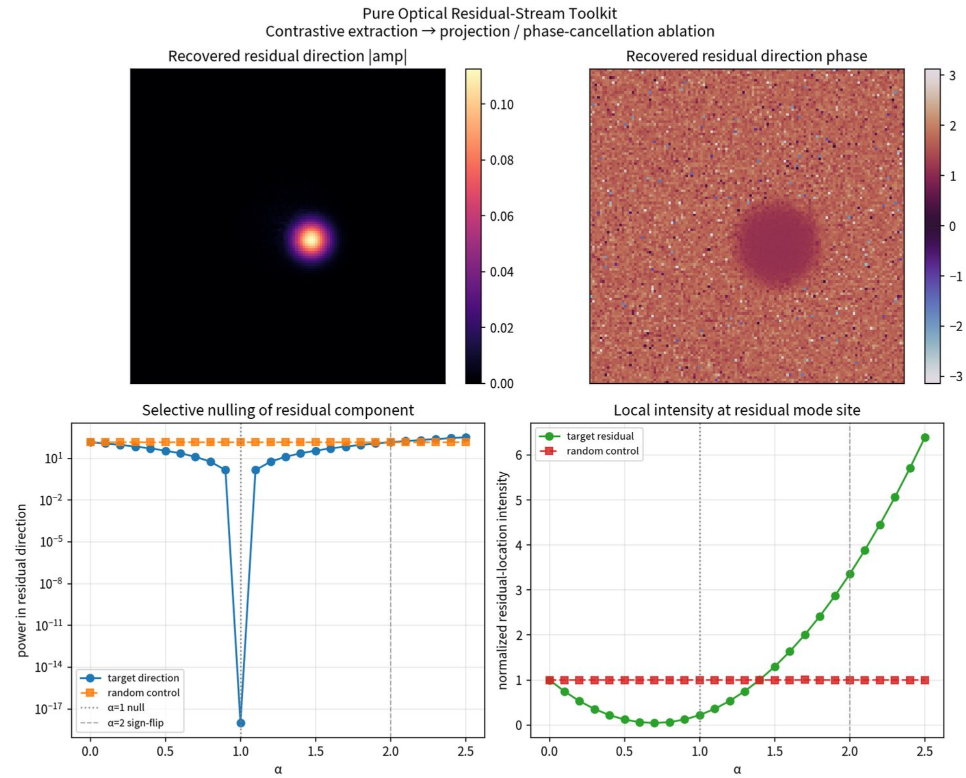

Figure 1. Static single-mode residual. Contrastive residual direction; residual power under projection / phase-cancellation; random control intact.

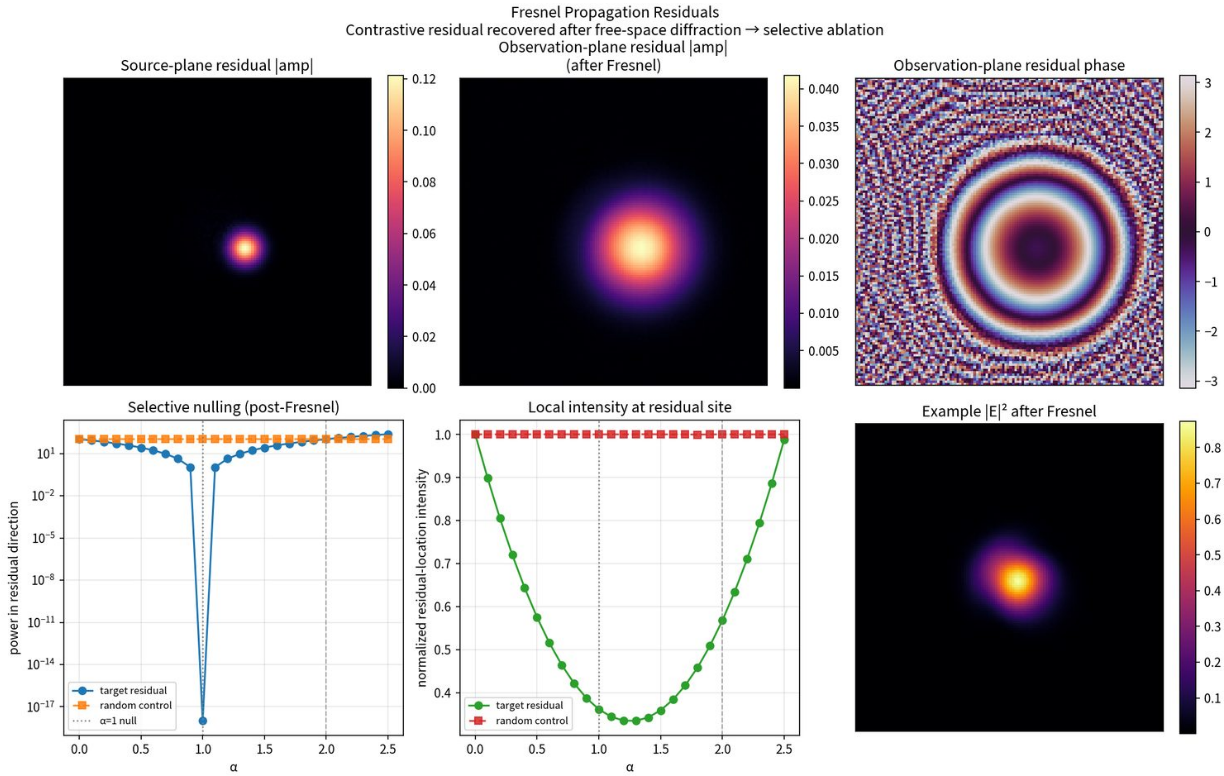

Figure 2. Fresnel-propagated residual. Residual remains selectively ablatable after free-space angular-spectrum propagation.

4.2 Multi-mode subspace and Jones polarization

Joint two-mode residuals were recovered by contrastive SVD. Top singular values captured the residual subspace; ablating the joint subspace collapses residual power at both residual locations; a matched random subspace does not. A vertical residual on a horizontal main beam was recovered with direction power fraction ≈ 96–99% E_y. Residual ablation collapses residual power and local |E_y|² while horizontal polarizer intensity remains stable.

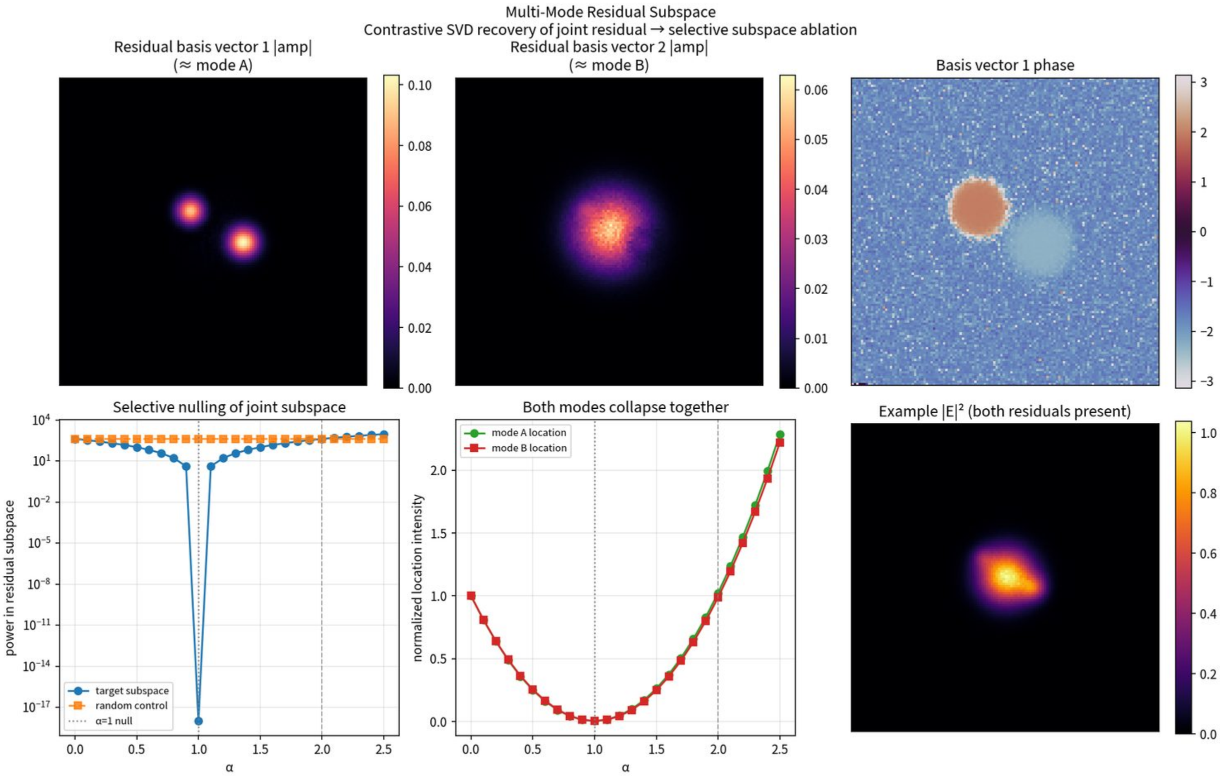

Figure 3. Multi-mode residual subspace recovery and selective joint-subspace ablation.

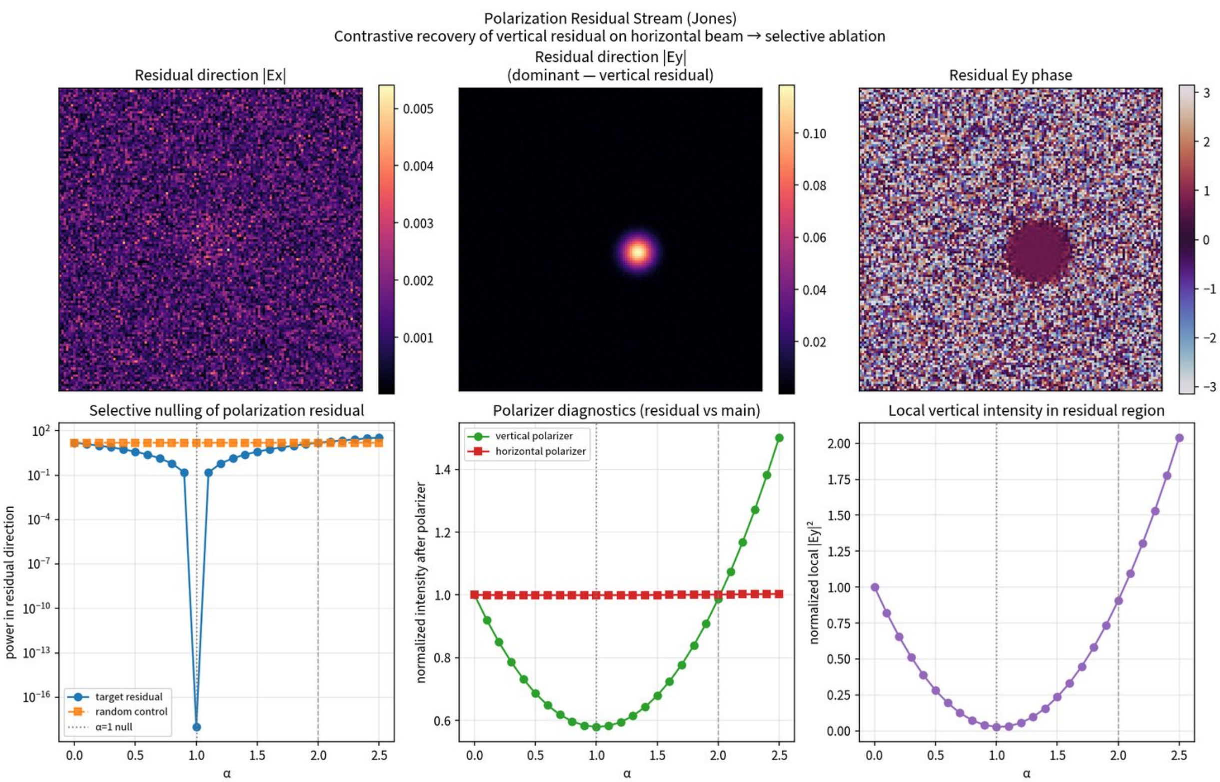

Figure 4. Jones polarization residual: vertical residual on a horizontal carrier; selective ablation with carrier protection.

4.3 Sparse dictionary recovery

Placing the contrastive residual direction as dictionary atom 0 and computing residual coefficients by projection yields selective separability: residual-atom ablation retains 0.0000% residual power; random-atom ablation retains ≈ 100%. The same pattern holds for multi-mode residual subspaces and polarization residuals.

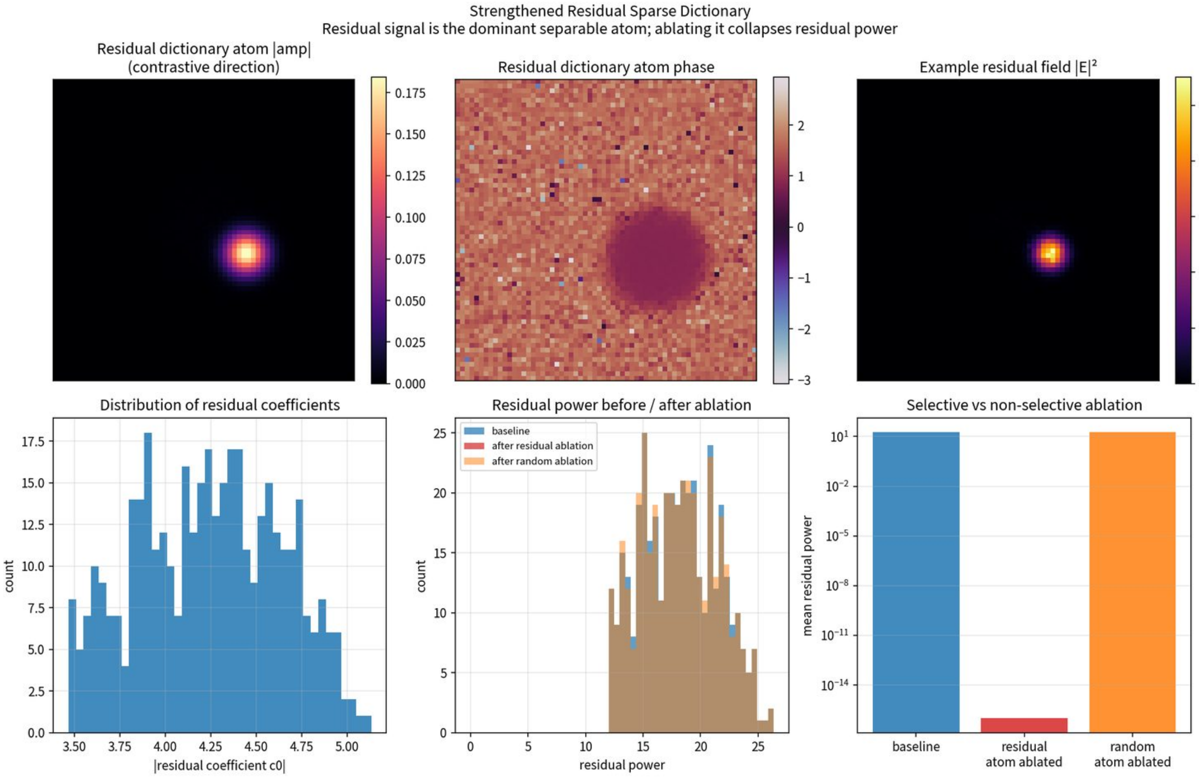

Figure 5. Sparse dictionary / residual-atom recovery. Residual atom matches the contrastive direction; residual-atom ablation collapses residual power.

4.4 Path-patch P1–P3

Single-mode (P1): residual power 53.31 → 15.95 (held-out field 16.11); local intensity 0.541 → 0.157 (held-out 0.158). Zero-patch collapses residual power; random-direction patch leaves residual power at 53.29. Multi-mode (P2): joint subspace path-patch moves subspace power and both local intensities toward held-out values. Polarization (P3): residual power 13.72 → 0.15 (held 0.13); local $|E_y|^2$ 0.168 → 0.001; horizontal polarizer intensity 0.04754 → 0.04755. P3 is the cleanest carrier-protection result in the optical suite.

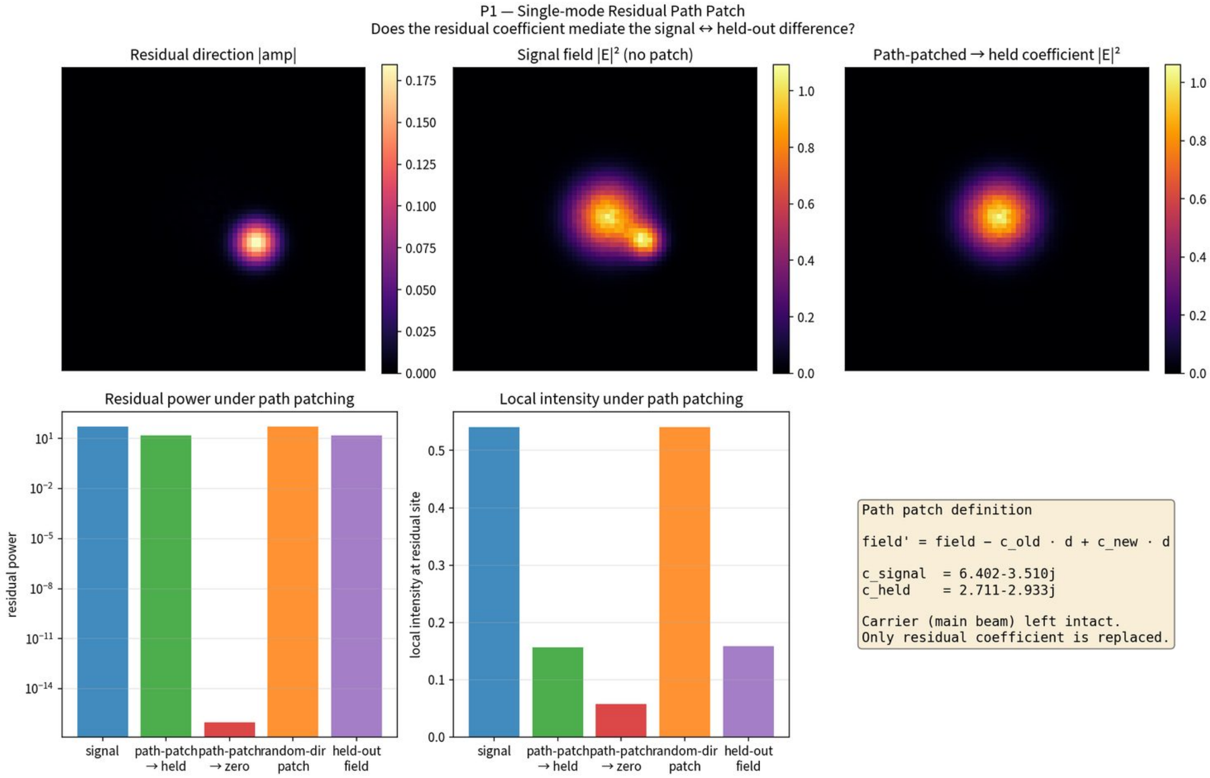

Figure 6. Path-patch P1 (single-mode): residual coefficient mediates residual power and local intensity.

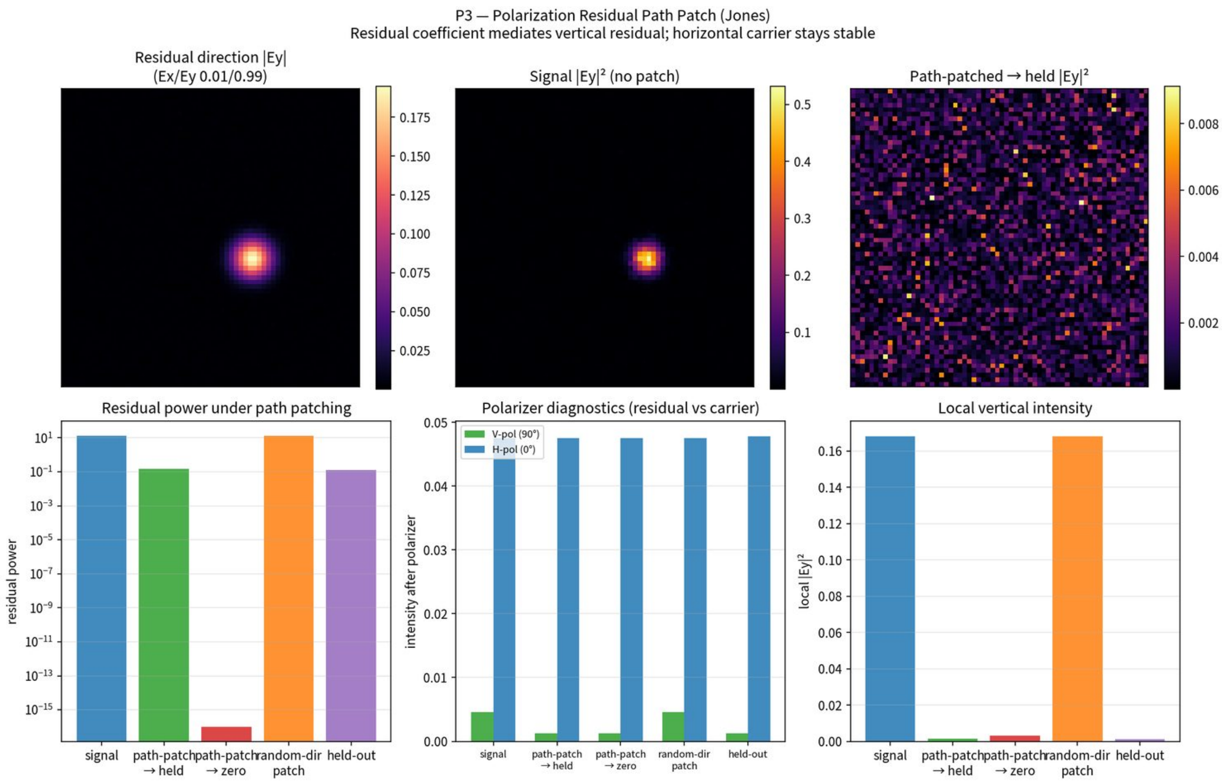

Figure 7. Path-patch P3 (Jones): residual coefficient mediation with horizontal carrier protection.

4.5 Residual Mueller matrices — synthetic and real

Residual diattenuator ΔM was recovered by contrastive least-squares with |cosine| ≈ 0.9997 to the true residual Mueller. Ablating residual Mueller (M → I) collapses residual Stokes power; random Mueller perturbations do not match residual-direction power. Residual linear retarder ΔM (phase residual) was recovered with |cosine| = 1.0000. The residual Mueller family thus covers both amplitude residual (diattenuator) and phase residual (retarder).

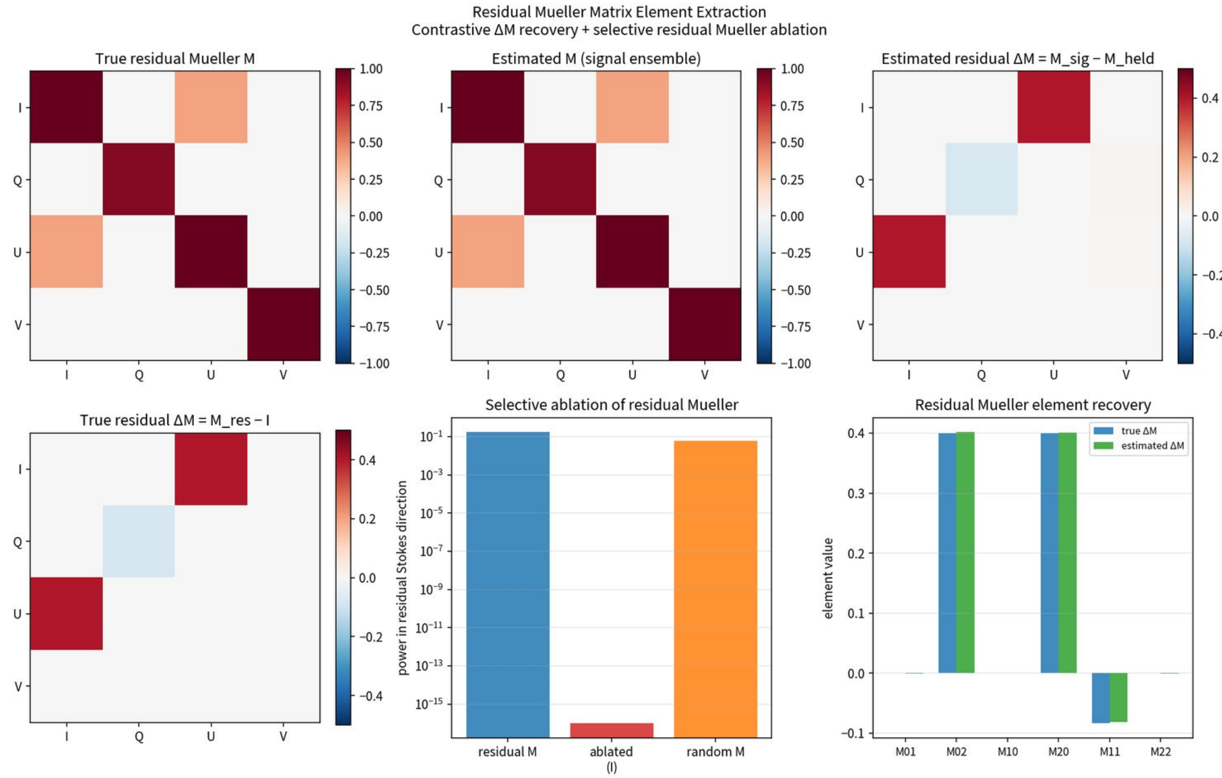

Figure 8. Residual Mueller diattenuator: contrastive ΔM recovery and selective residual-Mueller ablation.

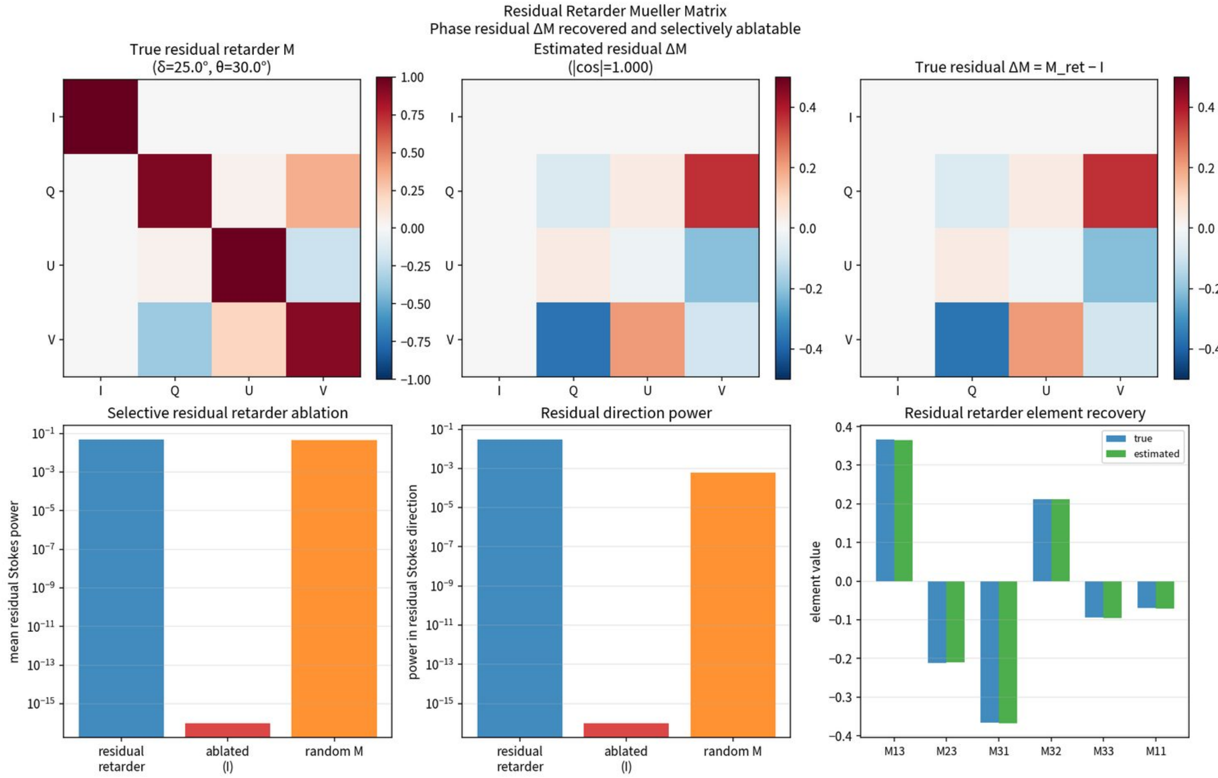

Figure 9. Residual Mueller retarder (phase residual). |cos| = 1.0000 to true ΔM; residual Stokes power collapses under ablation.

Public wavelength-dependent measured Mueller matrices from the UAH Mueller-Matrix Spectropolarimeter repository were used as measured residual systems relative to identity. Mean residual $\|M - I\|_F \approx 0.17$. Spectral residual SVD places ~96% variance in the top singular component. Element-wise residual ablation identifies load-bearing ΔM elements (especially M10, M22). Path-patch of residual structure between campaigns moves residual Stokes power (0.005 → 0.001). This is the first real (non-synthetic) optical residual Mueller result under the protocol.

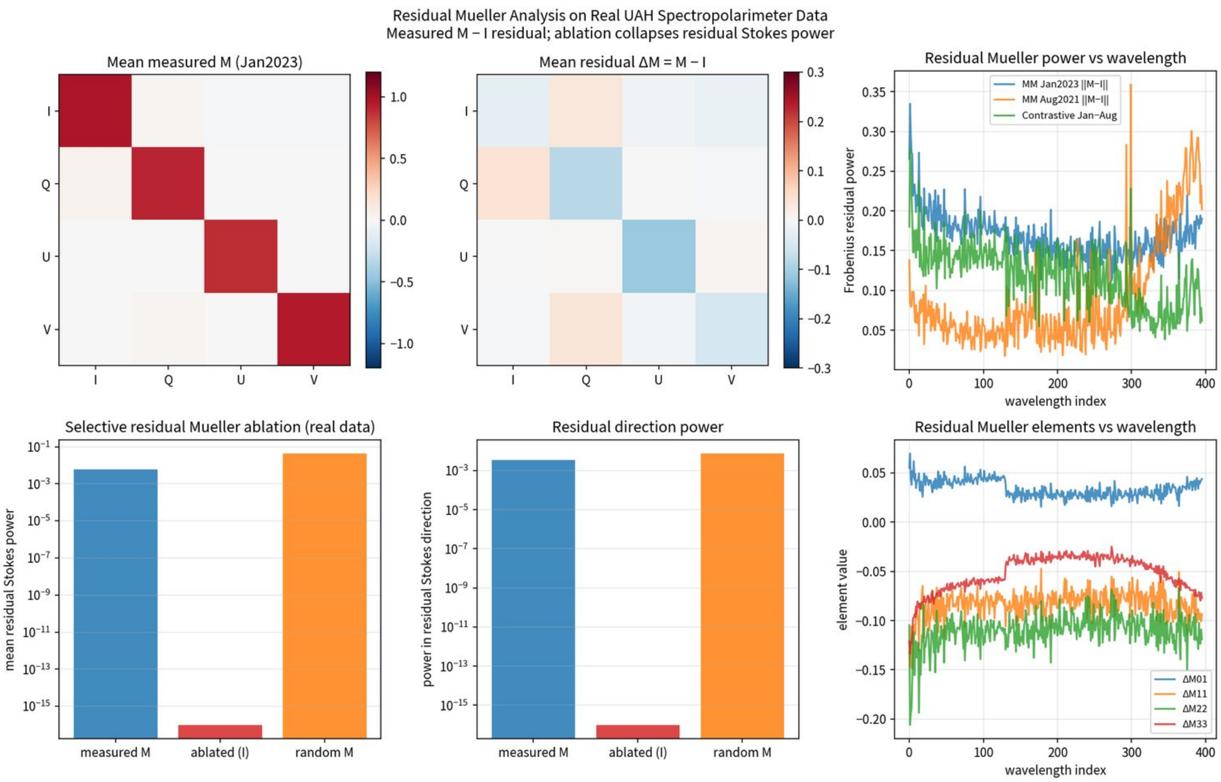

Figure 10. Real UAH spectropolarimeter residual Mueller. Measured MM treated as residual relative to identity; residual structure is low-rank and selectively ablatable.

5. Narrowband RF Residual Streams

Synthetic narrowband IQ mixtures with two carriers, three interferers, a memoryless cubic PA (producing classical IM3), and an additive residual tone were processed under the same protocol. Residual causal probe accuracy fell from 0.955 to 0.516 when the residual was zeroed; random controls stayed near 0.955. SAE recovery recovered all 17 physical features with mean |corr| = 0.922.

RF-P1 (single residual direction): residual power 320566 → 5.95 on path-patch to the held coefficient; zero-patch → 0; random-direction patch → 320544. Residual-frequency cosine ≈ 0.44; carriers / interferers cosine ≈ 0. RF-P2 (joint residual subspace, top-2 SVD, 61.5% variance): subspace power 612322 → 340; zero → 0; random subspace intact. RF-P3 checks IM3 parent-mediation: the residual direction recovers parent-driven variation that produces IM3, geometry consistent with physics. The primary gap — full coefficient path-patch on RF residual streams matching optical P1–P3 — is closed.

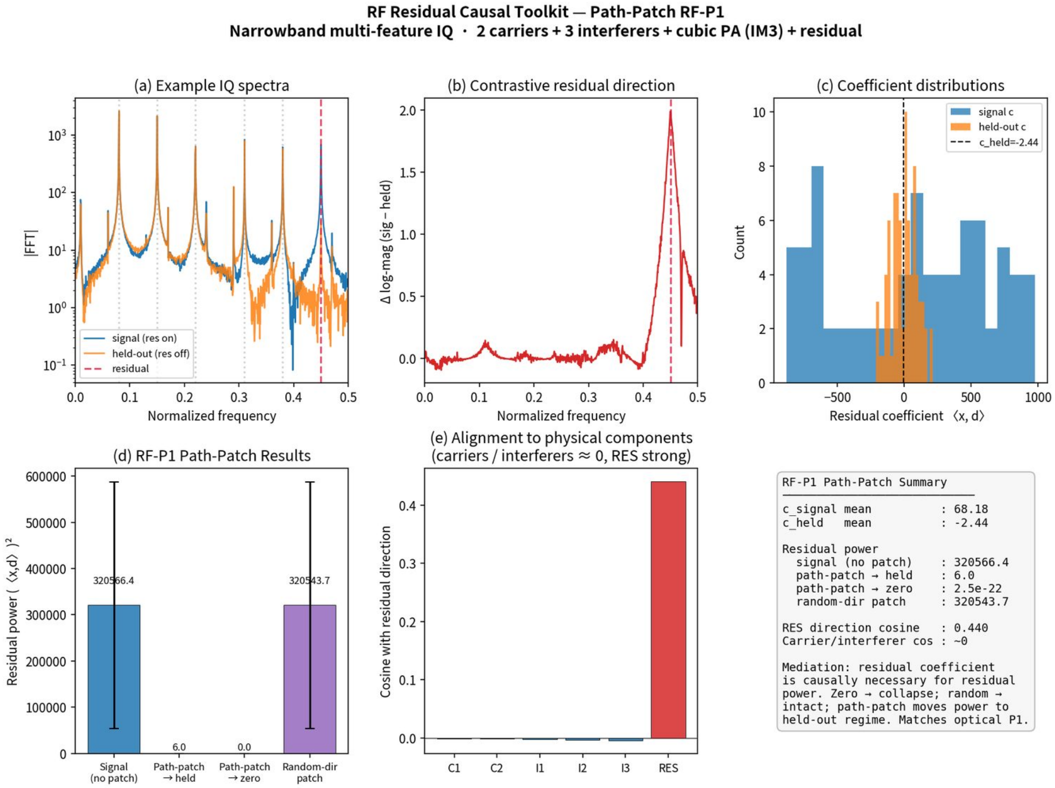

Figure 11. RF-P1 single residual path-patch on a synthetic narrowband IQ mixture.

6. Nitrogen-Vacancy Color-Center Residual Streams

Public single- and multi-NV ODMR data (Zenodo 10.5281/zenodo.14697917, Neu & Nimba epitaxial overgrown diamond) and a B-field ensemble (Figshare, 4693 sweeps) drop into the same pipeline. Temperature residual is constructed from the literature coefficient $\alpha = -74$ kHz/K.

Single residual (strain / E proxy): direction 99.999% along E; path-patch extreme Hole E = 94.8 MHz $\rightarrow$ 3.28 MHz (held mean); zero $\rightarrow$ 0; random intact; carrier D protected. Multi-NV residual subspace (dim-4): path-patch 0.315 $\rightarrow$ 0.0075. B-field residual: 0.00940 $\rightarrow$ 0.00055. Temperature residual: 1.59 $\rightarrow$ 0.85, with center frequency moving toward the held-out regime. Sparse atom ablation of the residual direction retains 0.0000% in all cases.

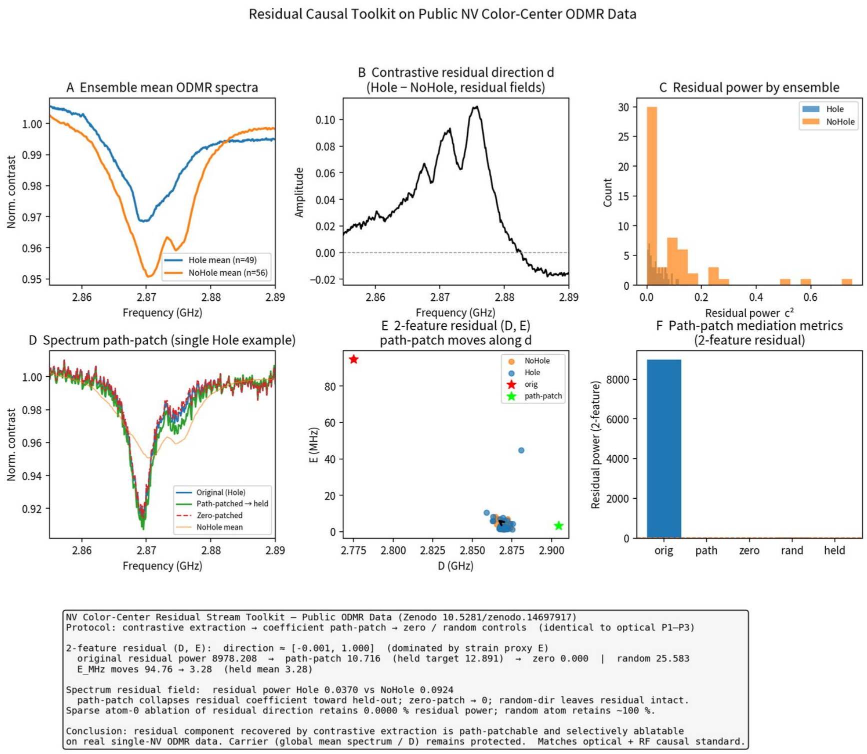

Figure 12. Single-NV ODMR residual toolkit on public Zenodo data. Residual direction is essentially pure strain-proxy E.

On the real Hole ($n = 49$) versus NoHole ($n = 56$) split, a 14-D pure-quadratic lift isolates nonlinear residual structure that the 4-D linear baseline misses: quad E^2 and prod $D \cdot E$ become causal ports. Residual-seeded SAE retains 0% residual power after residual-atom ablation in both 4-D and 14-D; 14-D recovers far higher residual power (~45.8 vs ~2.1) and concentrates ~88% of it into nonlinear features (max E^2 loading 0.777). Dominant SVD direction is essentially pure residual E^2.

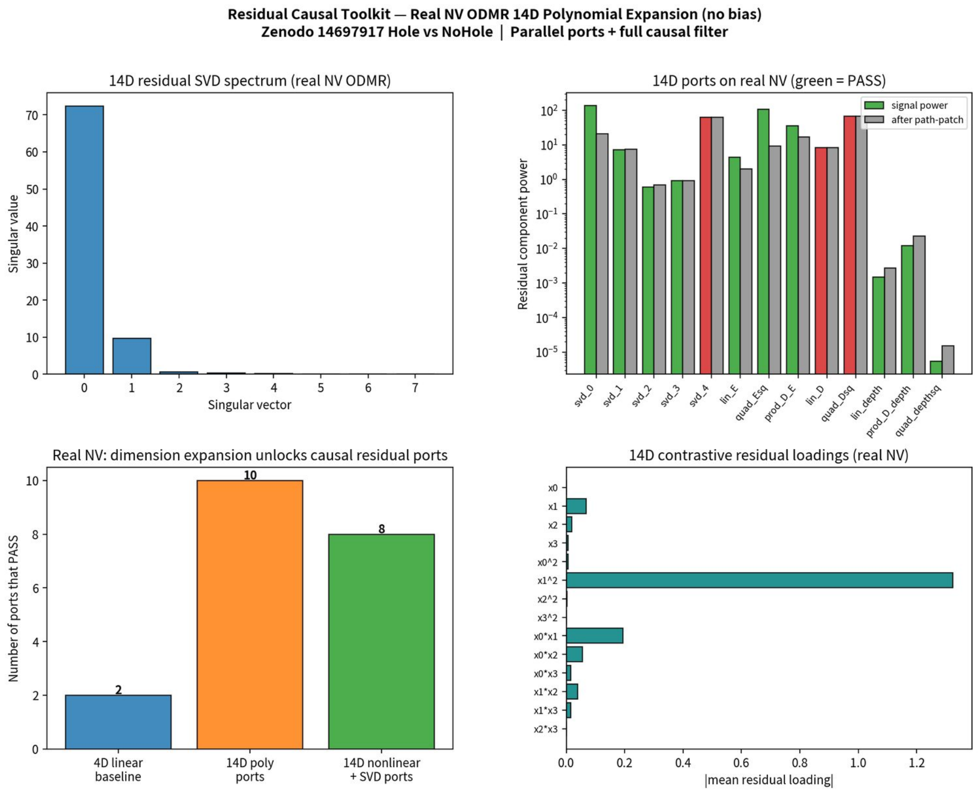

Figure 13. Real NV ODMR 14-D pure-quadratic residual ports. Nonlinear residual structure (E², D·E) becomes isolatable and causal.

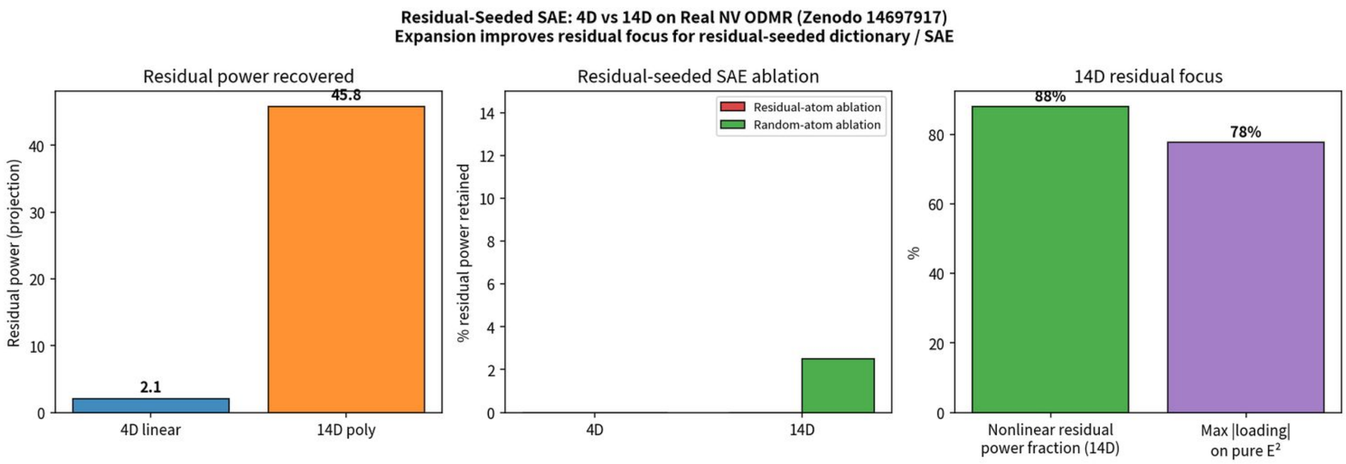

Figure 14. Residual-seeded SAE, 4-D versus 14-D, on real NV ODMR. Expansion improves residual focus; residual-atom ablation retains 0% in both cases.

7. Telemetry, EEG, CMB, and Audio Tones

7.1 Telemetry

Synthetic boat-style multi-channel mixture (NMEA engine RPM carrier ~2400 plus four vibration sensors; residual = additive mechanical harmonic near 3.3× rotational order): T-P1 residual-band energy 0.0448 → 0.0363 (exact held target); carrier-band protected. Real public naval propulsion telemetry (UCI Condition Based Maintenance of Naval Propulsion Plants, 11 934 samples × 16 GT sensor channels): residual = compressor performance decay; path-patch 1.374 → 0.0084; zero → 0; random-dir remains high (1.295); ship speed and GT rpm protected. Top residual features T2, T48, Pexh, TIC, GGn, mf — physically consistent with compressor decay.

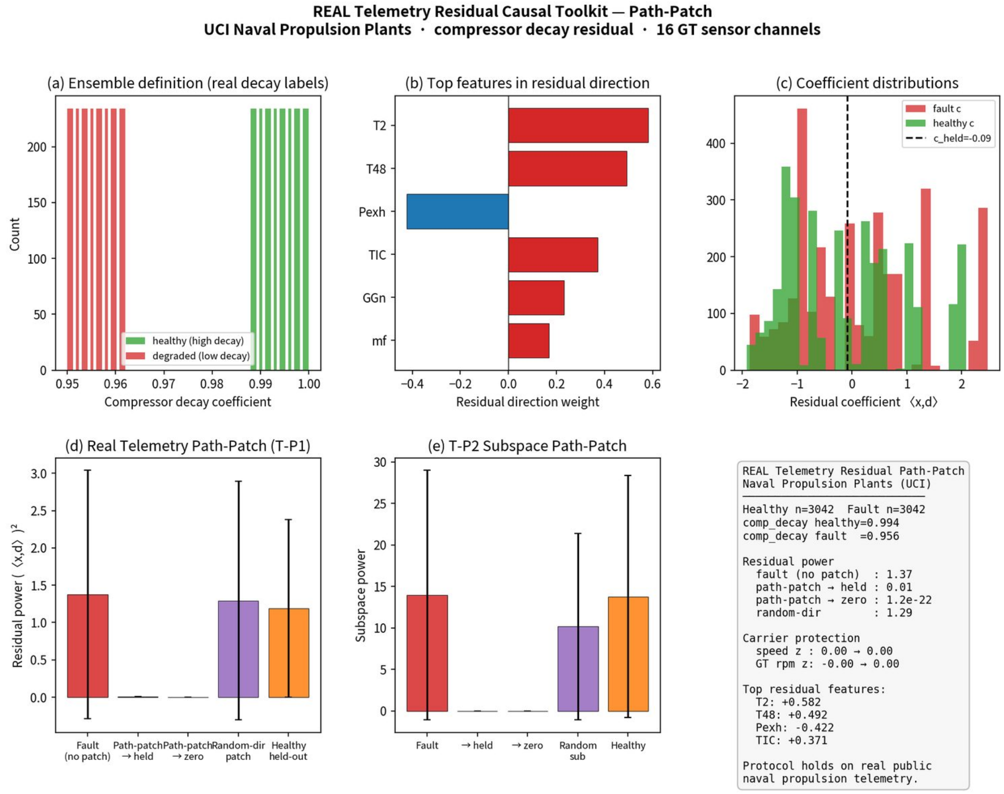

Figure 15. Real UCI naval propulsion residual. Residual = compressor decay; carriers protected.

7.2 EEG

Multi-channel question / comprehension residual stream: path-patch 66.3 → 1.35; residual-atom 0.0006% retained; random intact. A 5-D lift that appends an explicit differential / change coordinate recovers fleeting residual structure (path-patch 125.4 → 0.002). Real ZuCo / semantic-EEG features can drop into the same pipeline.

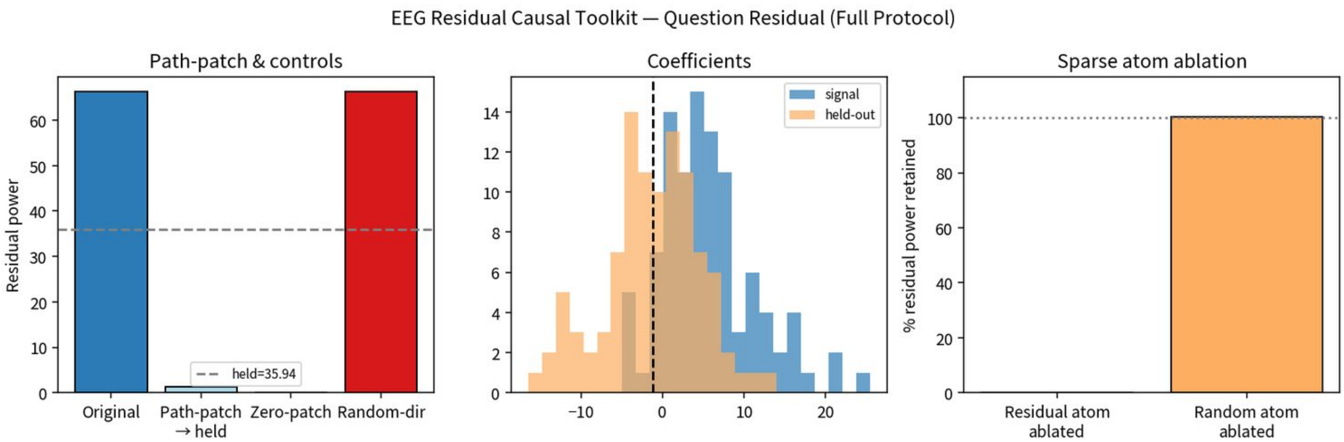

Figure 16. EEG question / comprehension residual. Full causal filter PASS.

7.3 Planck CMB residuals

Residual = data − best-fit ΛCDM (Planck R3.01 TT/TE/EE plus theory). Ports opened for TT, TE, EE, residual amplitude, and subspace velocity all PASS the causal filter. Residual-seeded SAE on those ports is selective but incomplete (residual-atom retains 36.76%). A 14-D lift finds a rank-1 residual whose dominant direction is residual-power² — the same second-moment pattern seen on NV (pure E^2) and RF (energy²).

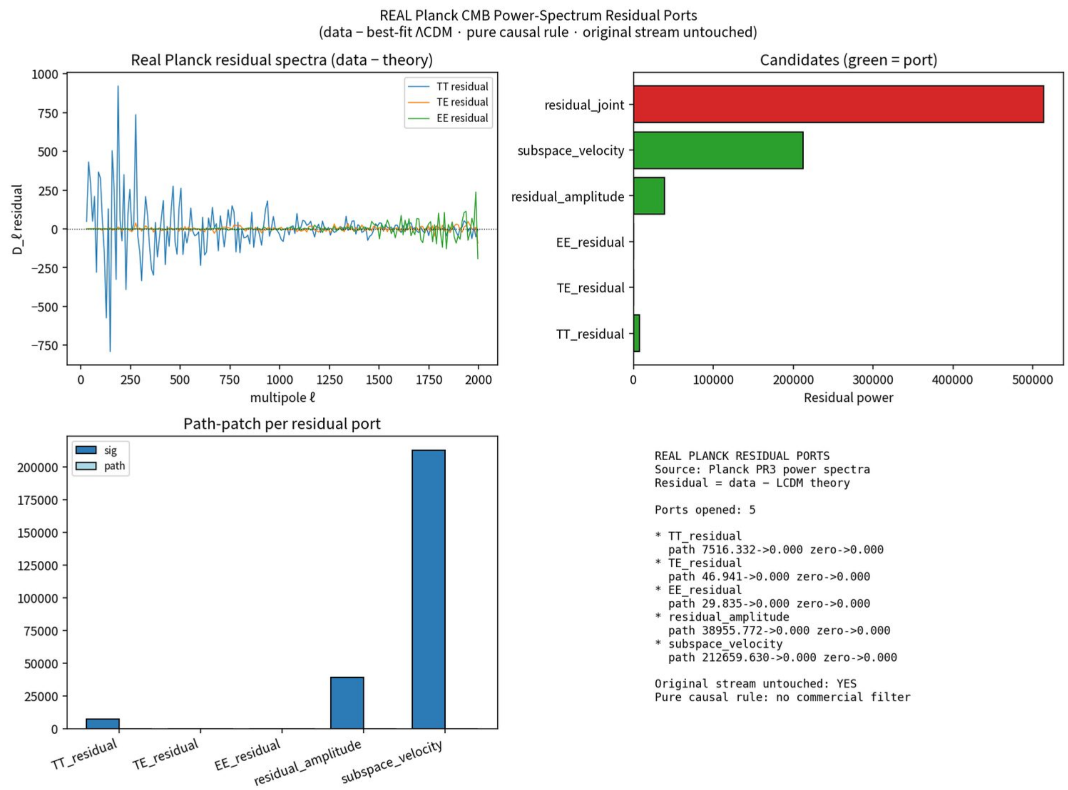

Figure 17. Real Planck CMB residual ports versus best-fit ΛCDM. Original stream untouched.

7.4 Audio residual tone

Controlled mixture with known components (fundamental 220 Hz + harmonics 440/660/880 Hz + residual feedback tone at 1250 Hz). Contrastive extraction recovers the residual direction peak exactly at 1250.0 Hz (error 0.0 Hz). Path-patch moves residual power 1018.09 → 0.100 (held target 0.100); zero → 0; random-direction → 1010.93.

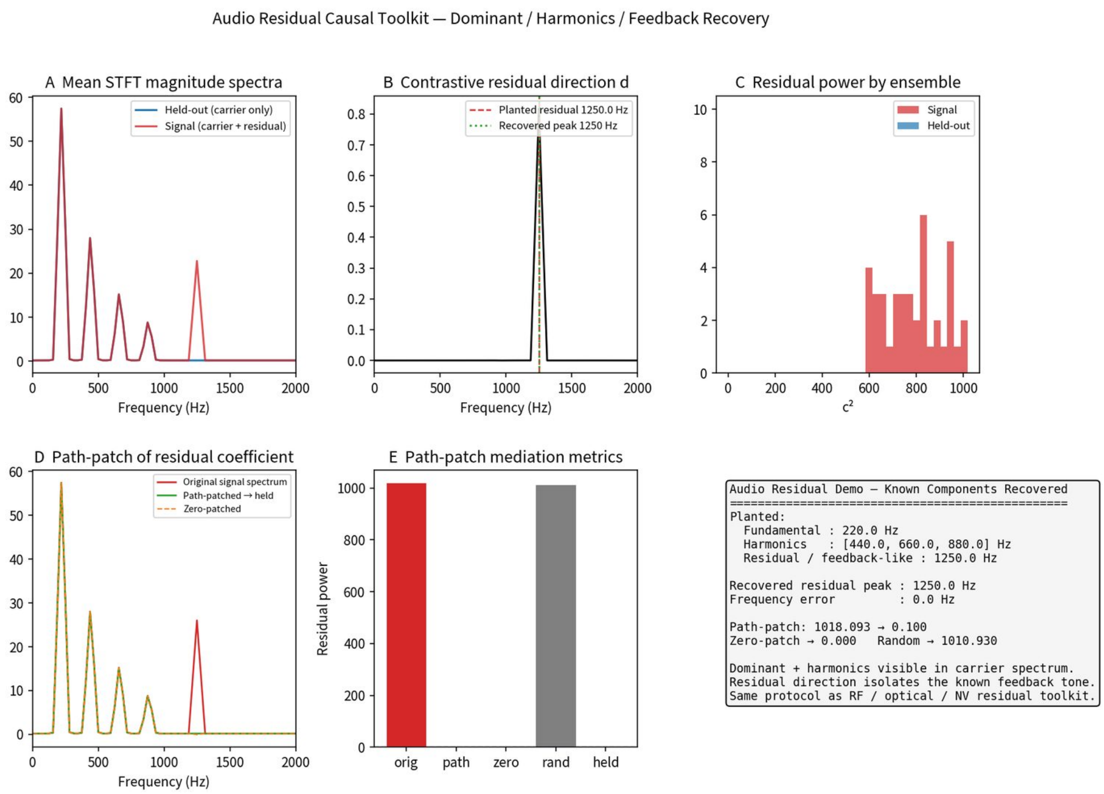

Figure 18. Audio STFT residual tone at 1250 Hz. Same protocol, same controls.

8. The 14-D Pure-Quadratic Campaign

Feature map: 4 linear + 4 pure squares + 6 cross products = 14-D, no bias, no centering. The same second-moment pattern appears across buses. Linear baselines recover the linear residual and miss the nonlinear ports. Dimension expansion surfaces additional causal nonlinear residual structure. Original 4-D streams remain untouched.

Domain	SVD0 variance	Dominant residual direction	Note
NV ODMR (real)	low-rank	pure residual E^2 (loading ~0.91–0.999)	prod D·E secondary
Planck CMB	~rank-1	residual-power² / Amp²	same 2nd-moment pattern
EEG	SVD0 ~97%	power² / maxabs²	residual power ~9× 4-D
RF IQ	SVD0 ~98%	pure residual energy² (~0.97)	power ~4685× 4-D
Modular arithmetic	SVD0 ~91%	residual r1² / feature² (~0.985)	computational bus
Tiny transformer	SVD0 ~88%	residual-power² (~0.95)	same pattern
Hybrid QNN residual	SVD0 ~97.6%	residual-magnitude-mode²	reconstructed Pauli Δr

Table 3. 14-D pure-quadratic campaign. Residual structure is low-rank; the dominant port is repeatedly a pure second-moment of the residual itself.

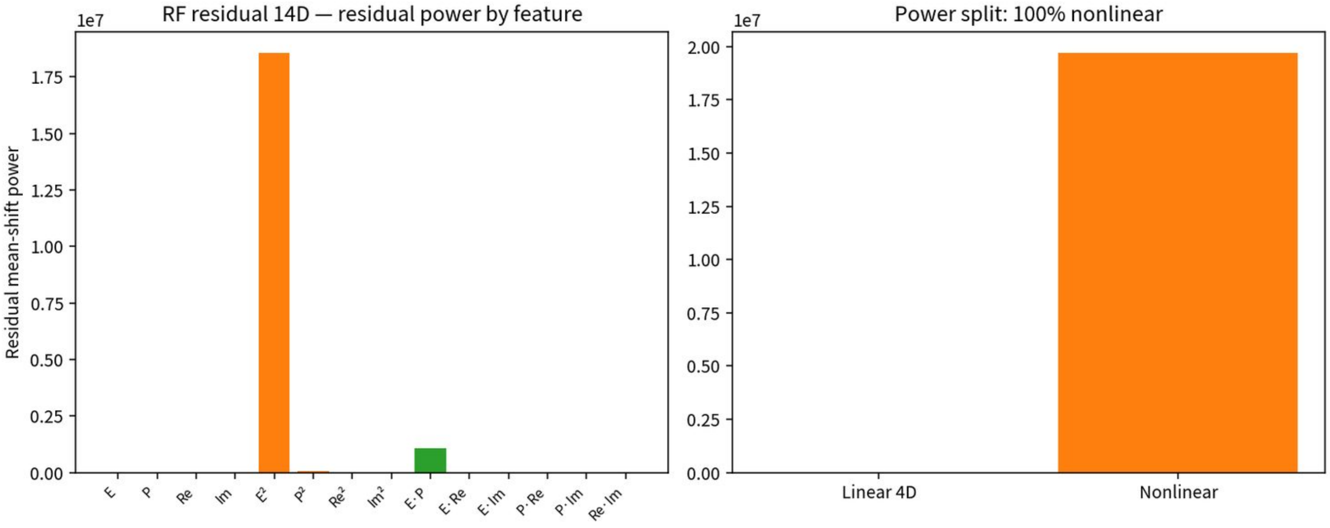

Figure 19. RF residual 14-D SVD. Dominant direction is residual energy².

The 14-D map is a fixed polynomial, not a learned dictionary. It is an extrapolation of the feature geometry, not a proof that nature is quadratic. What it shows is operational: before training an SAE, a cheap polynomial lift concentrates residual power into ports the causal filter can keep.

9. Parallel Ports and a Logical Residual Layer

The original stream is never modified. Parallel ports are opened only for components that pass the causal filter. A residual-seeded dictionary built from surviving ports can ablate joint residual power to 0%. Logical Residual Stream v0 takes multi-parallel observations (R1 linear 4-D, R2 pure-quad 14-D, R3 second-moment features) and forms a residual consensus layer. On real NV Hole/NoHole, 20/20 residual ports pass and consensus residual power 829.86 → 3.19 on path-patch, 0 on zero.

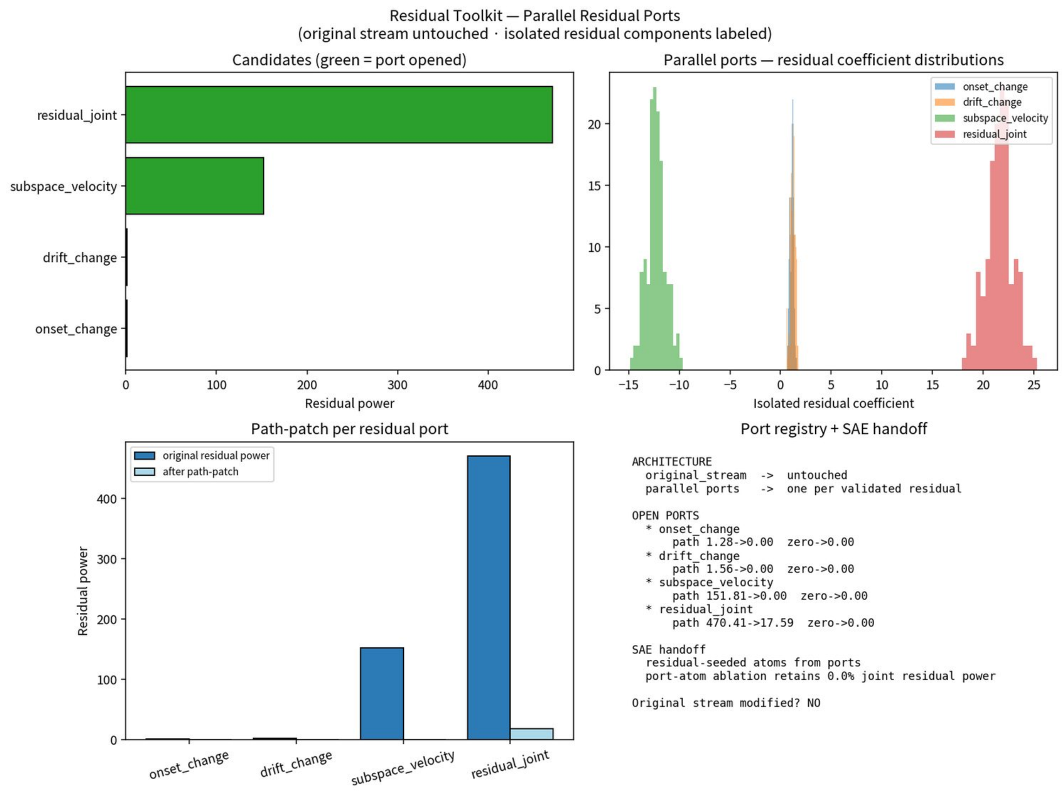

Figure 20. Parallel residual ports. Original stream S untouched; only FILTER-passing ports are opened.

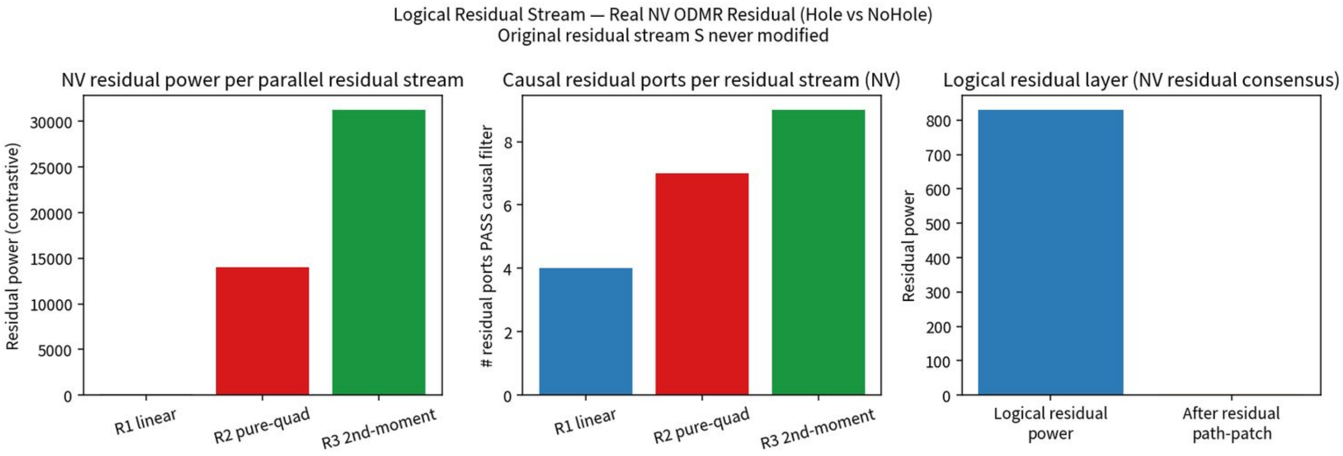

Figure 21. Logical residual stream on real NV ODMR. Consensus residual power collapses under path-patch.

10. Quantum Residual Streams

A companion experiment treats the mid-circuit state of a 4-qubit hybrid variational circuit as a residual stream [Taddonio, 2026b]. After training, a frozen intervention — either a pure phase rotation RZ(θ) or an amplitude-mixing rotation RY(θ) — is inserted and swept. Phase ablation produces a robust, non-monotonic causal response across 32 random seeds (mean max accuracy swing ≈ 0.23; 56% of seeds exceed a 0.20 drop). Extracted Pauli residual vectors show a structured pre/post difference whose magnitude tracks performance with correlation r ≈ 0.97 and is low-rank (top-3 principal components explain ≈ 75% of residual-shift variance). Every qubit is load-bearing under both phase and magnitude interventions; phase sensitivity increases toward later qubits; magnitude effects remain strong and comparatively uniform.

Qubit	Phase RZ mean ΔAcc	Magnitude RY mean ΔAcc
q0	0.204	0.318

Qubit	Phase RZ mean ΔAcc	Magnitude RY mean ΔAcc
q1	0.254	0.302
q2	0.267	0.293
q3	0.333	0.328

Table 4. Hybrid QNN multi-qubit ablation map (12 seeds per cell). Every qubit is load-bearing.

A 14-D lift on residual ensembles reconstructed from that Pauli geometry again finds a low-rank residual (SVD0 ~97.6%) whose dominant direction is residual-magnitude-mode². Caveat, locked: live QNN residual activations were not re-run from a stored dump; the 14-D quantum result uses reconstructed residual geometry consistent with the published Pauli residual paper. Scale is small (4 qubits, statevector, synthetic labels). The claim is transfer of method, not of large-model phenomenology.

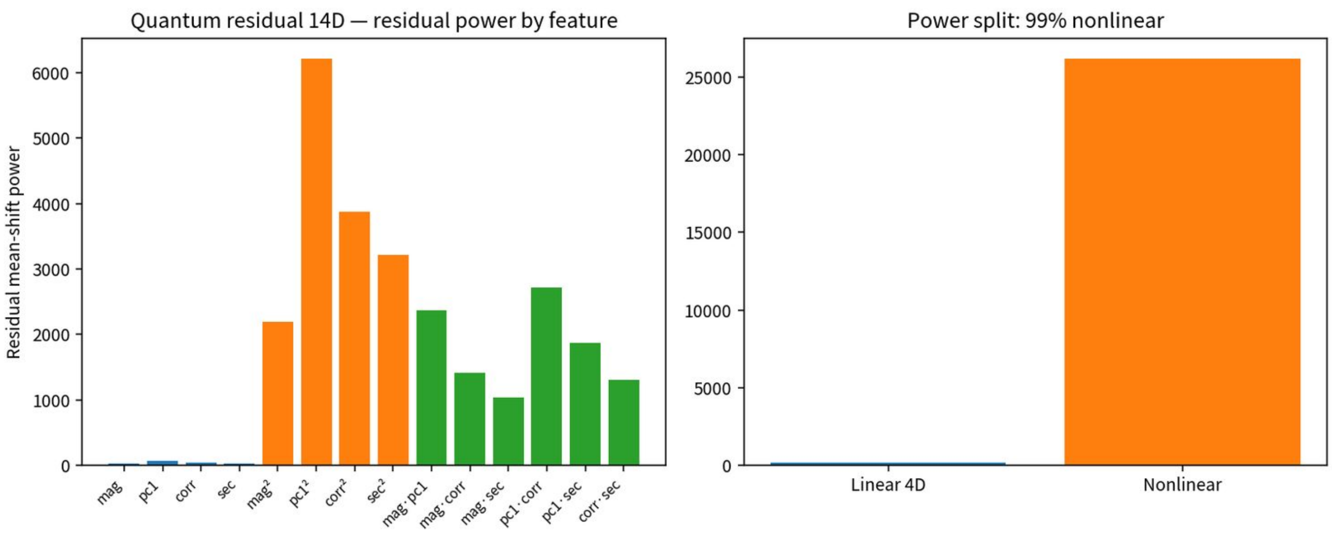

Figure 22. Reconstructed hybrid-QNN residual 14-D SVD. Same second-moment residual pattern.

The parallel with the classical residual stream is structural. Intervention axes (phase rotation vs RZ; magnitude scale vs RY), causal signature (accuracy collapse vs accuracy swing), low-rank residual structure, and a tight performance link (phase cancel ⇒ chance classically; ■Δr■ correlates $r \approx 0.97$ with drop here) recur on a quantum state. Residual causal geometry is not confined to classical activations.

11. GPT-2 Phases A–D: Offline Filter to Online Write-Hook

Modular-arithmetic GPT-2, layer 6. Phase A froze residual ports (R1 linear 4-D / R2 pure-quad 14-D / R3 second-moment); n_pass mean ~23.6–24.5 / 27, S untouched on all seeds. Phase B tracked the residual trajectory during training. Phase C ran an offline residual closed-loop on high-power R2_svd_0; 8/8 seeds causal_pass + control_success + zero_collapse + random_intact.

Phase D is the non-redundant next claim: an online write-hook during the forward pass, resid ← resid − c·d + c_target·d at the last position. Sixteen seeds. Zero collapses 16/16; coefficient matches held 16/16. Mean Δacc: zero −16.1%, path −22.0%, random −0.2%. Mean Δlogprob: zero −0.75, path −1.56, random −0.001. The residual coefficient mediates answer probability selectively. This is behavior control, not only residual-power control.

Phase D-stronger lifts R2_svd_0 into the write-hook via shared-PCA → 14-D → ridge residual-to-coefficient mapping and is smoke-validated. Online write-hooks change behavior on modular GPT-2. That is not yet a general behavior-control result on language models trained on natural text.

12. Residual Interferometer

A unitary 2-D core mix PHASE(ϕ) + BEAM(θ) was built as the first closed ISA demonstration. Vector-core fringe contrast 0.788 versus random 0.303; optical 2-mode peak-to-peak 298.4 versus random 0.0001; unitary True; complement untouched. Coefficient = 1 leaks 32% energy; coefficient = 2 does not. Phase E was packaged for a 4×4090 16-seed online PHASE(ϕ) on the GPT-2 2-D core.

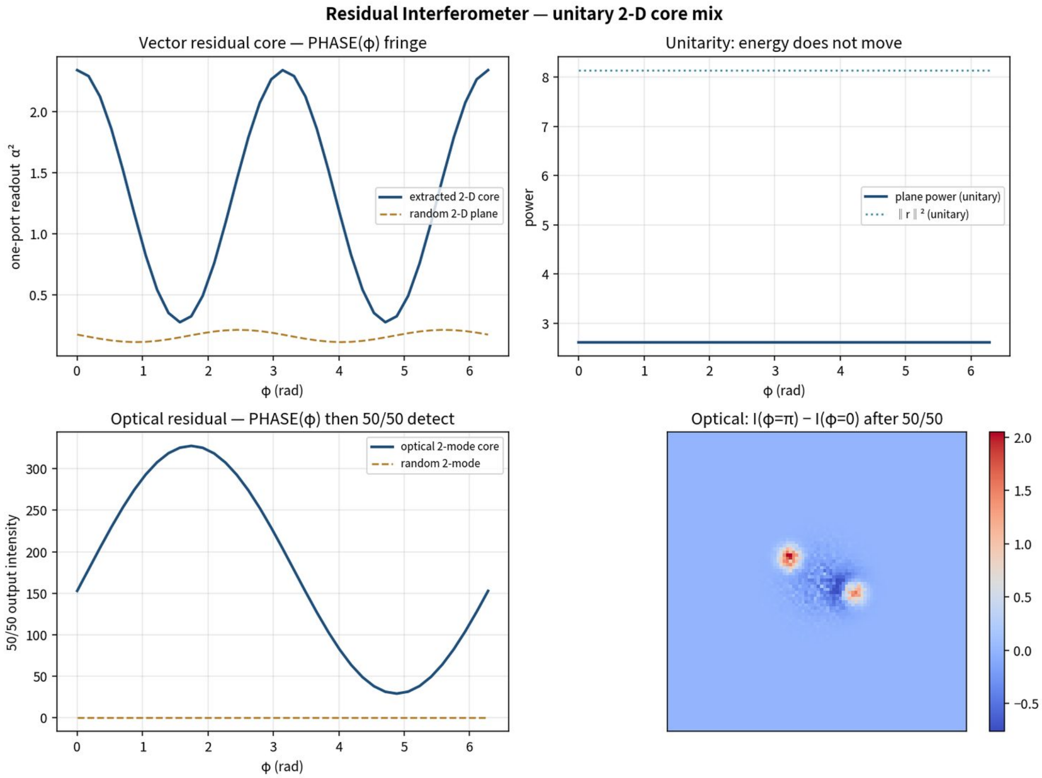

Figure 23. Residual interferometer. PHASE(ϕ)+BEAM(θ) on a legal 2-D core; random plane is the control.

13. CRAI: Causal Audio Isolation

CRAI began as Reference-Driven Causal Audio Isolation and was corrected under a professional isolation bar. CRAI-1 used a B-gated contrastive plus exact 14-D map. Waveform PHASE($\phi = \pi$) cut hit-frame crack / air leakage 97.13% with 99.95% carrier retain. Operational contrast is B-loud versus B-quiet frames of A — literal mean(A) − mean(B) isolates the carrier, not the bleed.

CRAI-2 dropped the live reference-mic oracle. Time / phase comes from source-mic distance. Mixture-domain suppress was not printable (guitar SI-SDR 11.95 → 13.52 dB; isolated guitar still +21 dB crack versus clean).

CRAI-3 rebuilt the output from the carrier port, not from mixture holes. Guitar: additive partial resynthesis (f0 = 196 Hz, median envelopes, phase fit on non-hit frames), SI-SDR 11.95 → 23.96 dB, crack −29.6 dB versus raw and matching the clean source. Snare: envelope-delay of isolated guitar (6.19 versus planted 6.20 ms) then subtract; H2/H3 bleed −25 dB, SI-SDR −3.74 → 19.58 dB. STATUS PASS on the professional bar for this synthetic scene. Real guitar will need a richer carrier model.

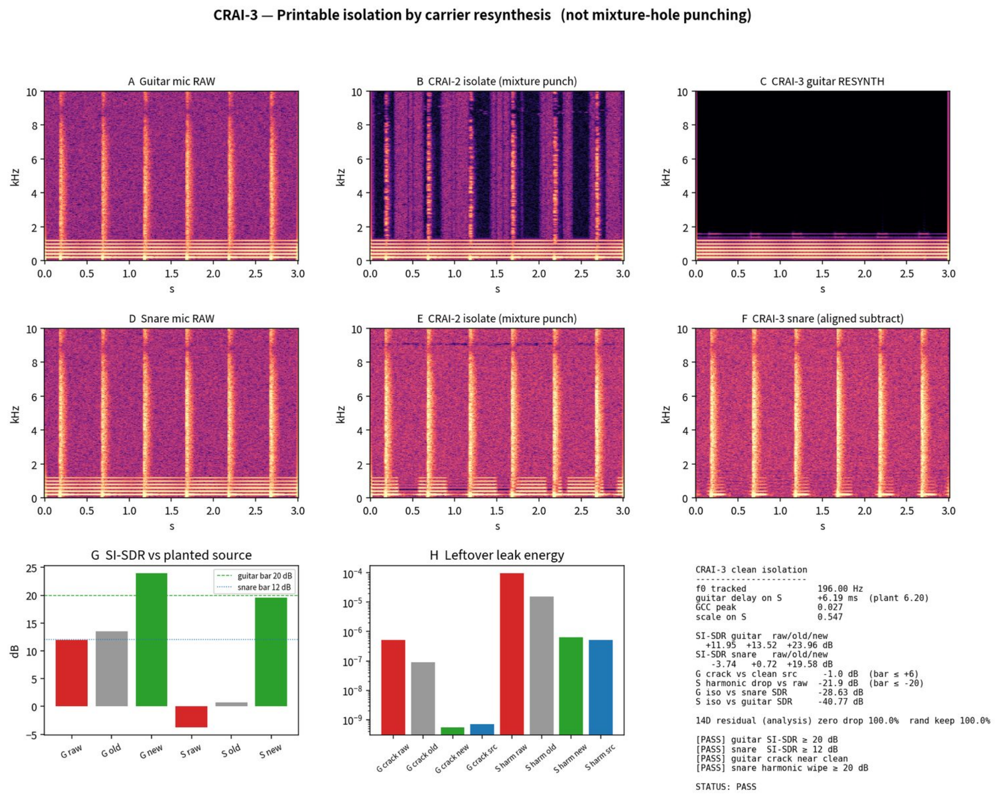

Figure 24. CRAI-3 clean isolation. Output rebuilt from the carrier port. Synthetic scene PASS on the trained-ear bar; computer metrics are supporting evidence only.

Product lock: toolkit DSP only — no neural net, no weights, no GPU inference. Single-insert drop-in is valid; isolation does not need a reference / sidechain mic. 96 kHz live constraint: STFT insert latency is illegal. Causal lock-in bank at 96 kHz, throughput 0.333 ms at a 32-sample block. Isolation depth is below CRAI-3 offline, as required by causality.

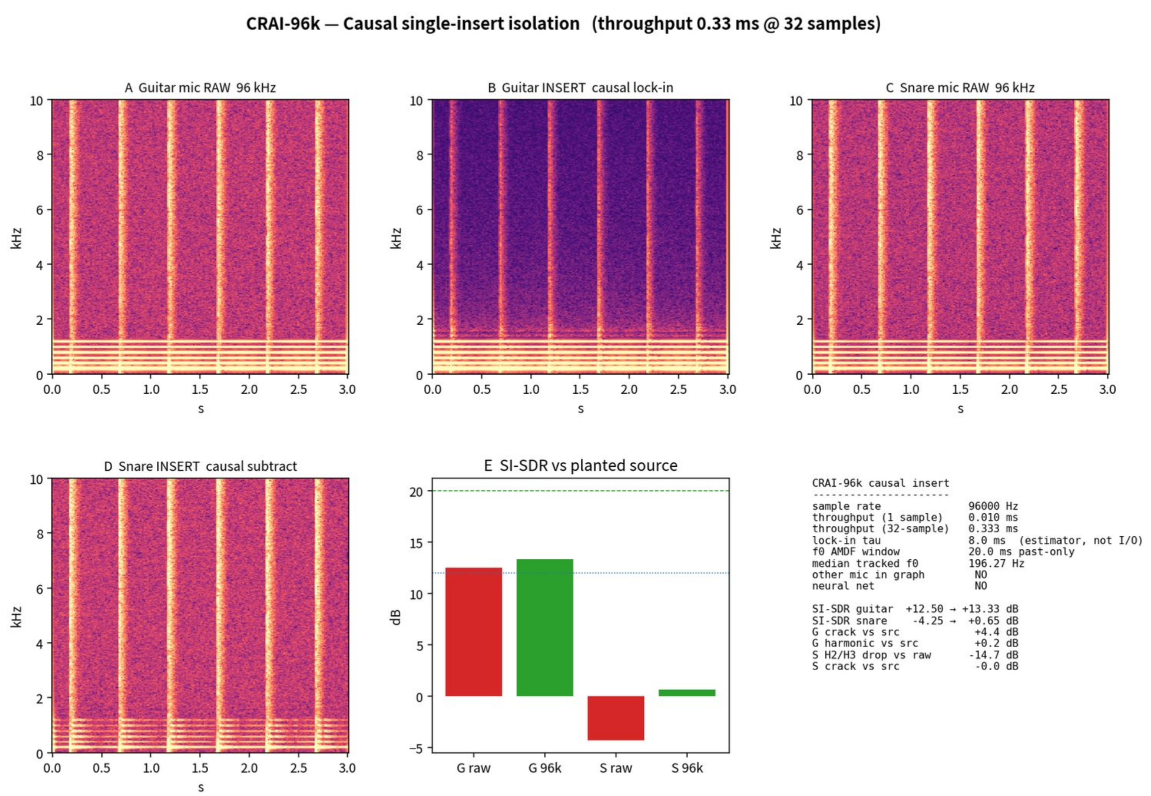

Figure 25. 96 kHz causal lock-in isolator. Legal latency; weaker isolation depth than offline CRAI-3.

14. Continuum Processor P1–P3

An FNO-form processor G(u) = W u + F■^{1}(R(k) F(u)) was placed on an optical field bus. No neural-operator training. The question was whether FILTER still holds after a linear field processor, and whether PHASE can be written on the source plane and read after G.

P1: address A output-field port PASS (115.30 → 7.36 path / ~0 zero / 115.29 random). Address B residual-multiplier band ΔR PASS. |cos|(ΔR, planted) = 1. P2: joint 2-D core. Co-occurrence SVD is rank-1 (svd0 = 0.989); legal core is two single-feature probe arms, not SVD-1. Joint FILTER PASS. PHASE + BEAM(π/4) one-port ptp 290.90 versus random 0.014. BEAM(π/2) arm-swap true. P3: source-plane PHASE write-hook, READ after G. Source arms must be orthogonalized against the carrier or hard-zero punches holes that G diffracts back into the residual READ. FILTER PASS. Source plane / complement / total flat to 1×10■^{1}■. Output-plane power is not flat in ϕ — G(carrier) interferes with the phased residual after the linear operator.

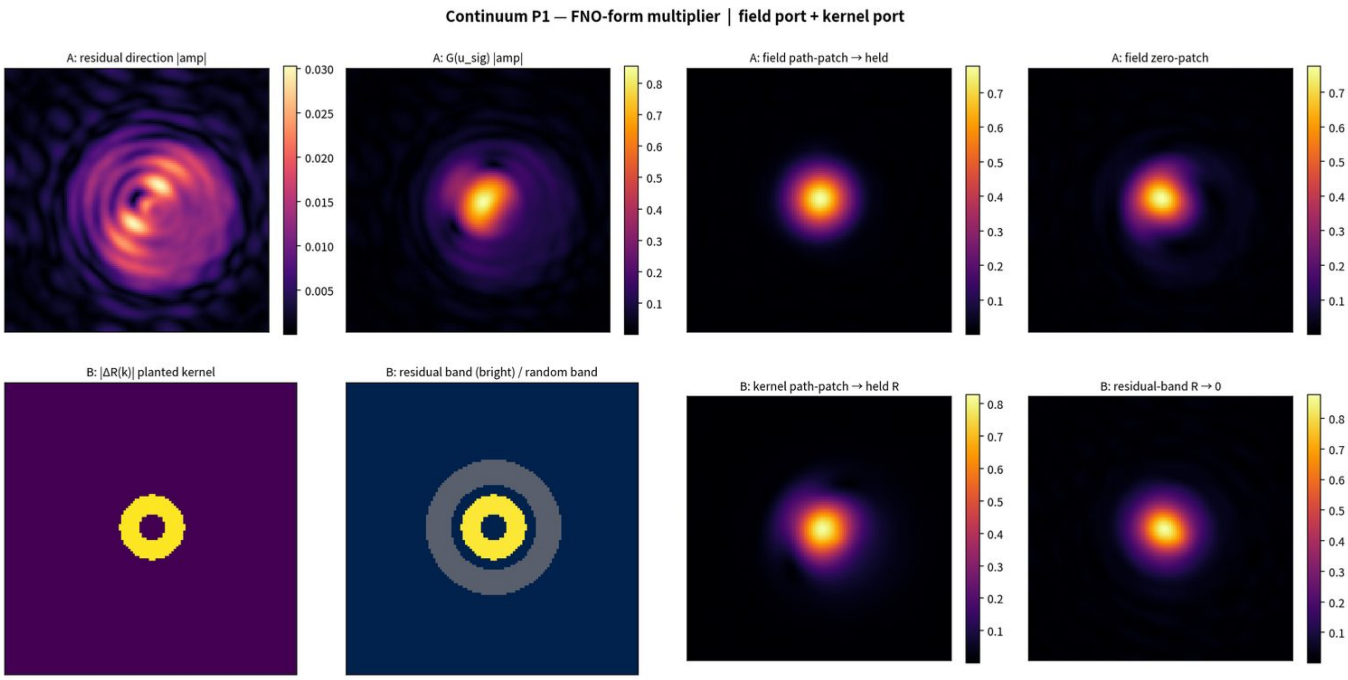

Figure 26. Continuum P1. Field port and kernel-multiplier port ΔR on an FNO-form processor.

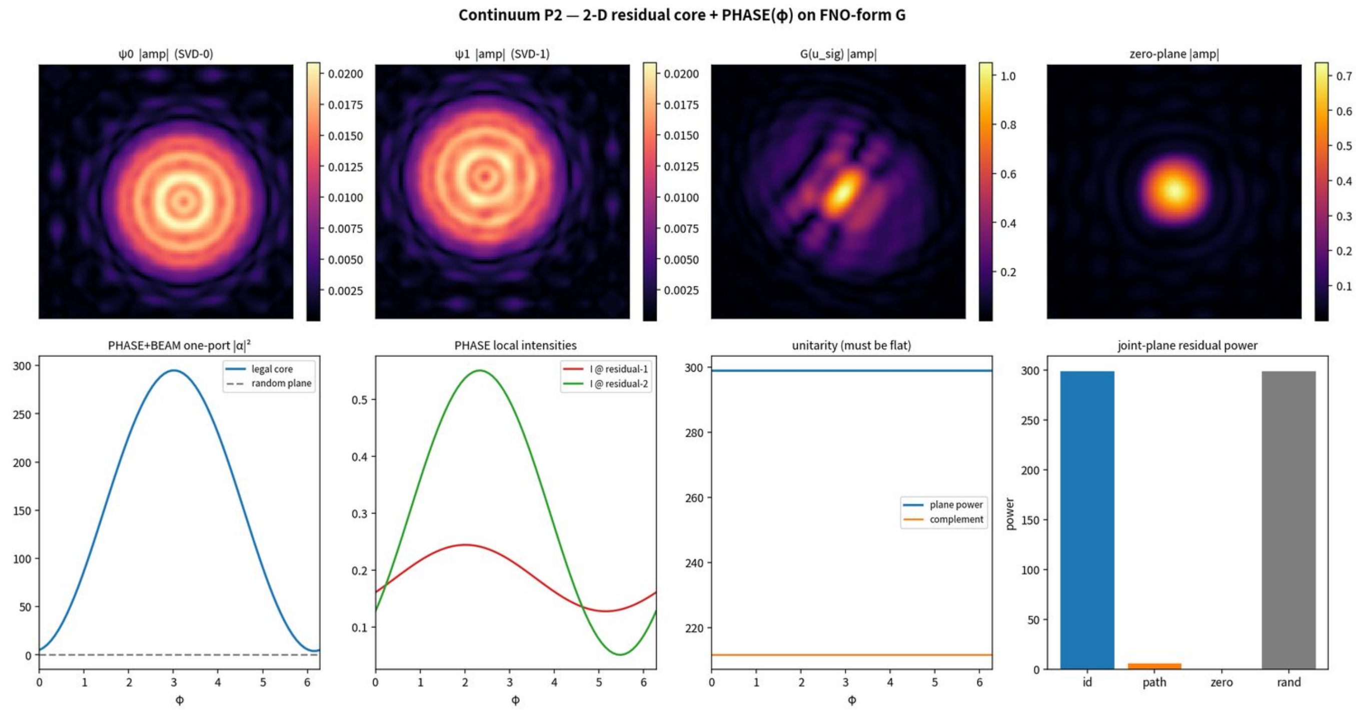

Figure 27. Continuum P2. Legal 2-D core from single-feature probe arms; co-occurrence SVD is rank-1.

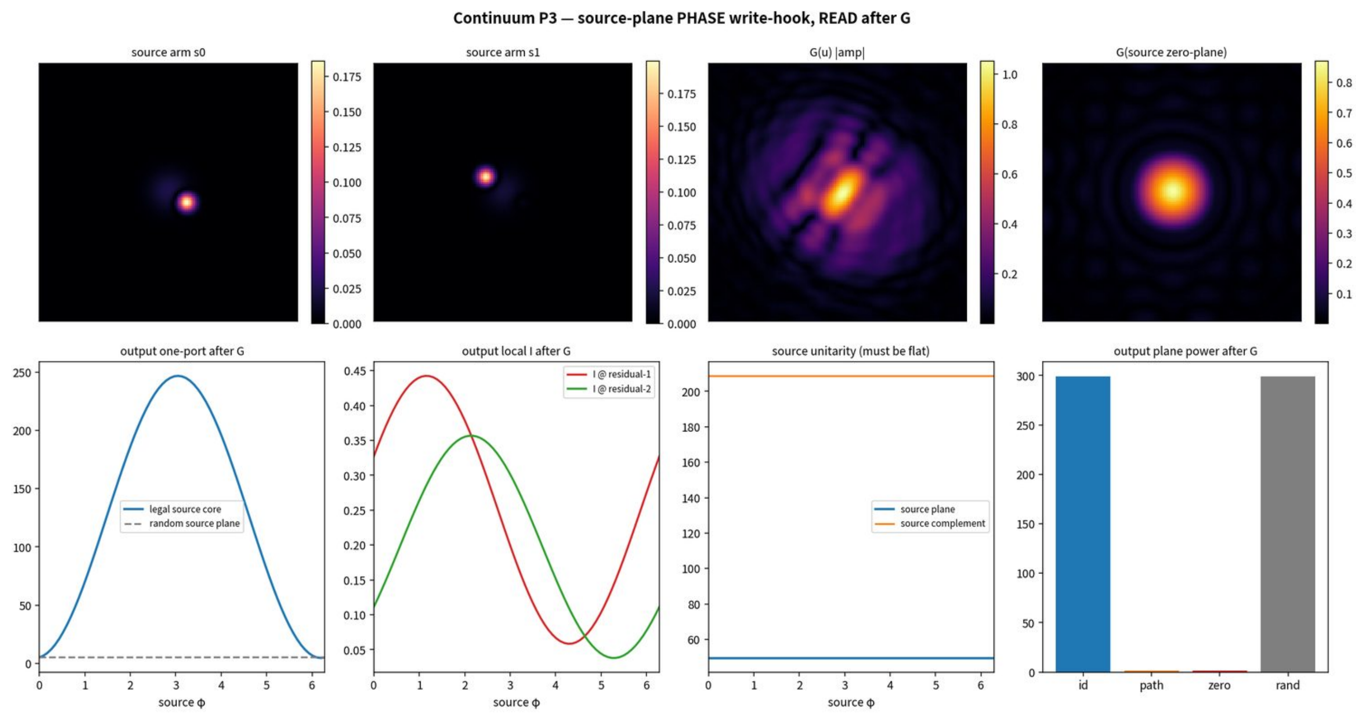

Figure 28. Continuum P3. Source-plane PHASE write-hook, READ after G. Source unitarity flat; output-plane power breathes.

Decision recorded 31 August 2026: a neural operator is a processor on the Residual Stream ISA, not a replacement ISA. Relates to Accelerated Understanding (Anandkumar / Jenik, launched 25 August 2026) as the causal-address / directional-feedback layer their 4-D simulator does not itself provide.

15. Trainer Test and Owned RF Plant

15.1 Train A/B — honest FAIL

Same architecture, two arms, 8 seeds × 2 arms on a 4×5090 box. GPT-2 / mod-113 / layer 6 / 6 epochs / n_train = 4096. Arm A: cross-entropy only + FILTER eval. Arm B: cross-entropy plus a residual READ in-loop, same FILTER.

Arm	Acc	FILTER	p_sig	p_path	Epochs-pass
A (CE only)	0.224 ± 0.045	4/8	245.5	108.1	3.00
B (CE + residual READ)	0.185 ± 0.050	3/8	17.73	8.94	2.12
Δ B − A	−0.039	−1	−227.8	—	—

Table 5. Train A/B. Success rule FAIL: B is not ≥ A. The toolkit is not claimed as a trainer on this task.

Sidecar as written used τ_sig = 1.0 while observed p_sig is O(100–400), so Arm B's held / complement terms flatten the port. Accuracy on both arms is well below the Phase D 36.2% baseline (undertrained relative to that lock). Next legal move on this axis is a same-architecture sidecar retune of τ and lambdas — not a third architecture, not a neural-operator stack.

15.2 RTL-SDR Blog V4

RTL-SDR Blog V4 (R828D + RTL2832U, 1 ppm TCXO, HF bias tee, SMA, dipole kit) arrived 3 September 2026. Receive-only. Stick enumerates on Windows (Zadig WinUSB + Blog V4 rtlsdr.dll). First real IQ dump 7 September 2026: fm_sanity.cu8 = 8 192 000 bytes (4 096 000 CU8 samples) at LO = 97.5 MHz, fs = 2.048e6, gain = 20.7 dB. Hardware path PASS. Preferred first plant: conducted coax + pads, SDR as RX only, separate software-controlled interferer TX, mute / gain actuator. Closed loop = READ residual port → FILTER → WRITE an actuator. Received S stays untouched. No live TX rewrite until FILTER = PASS on files this radio wrote.

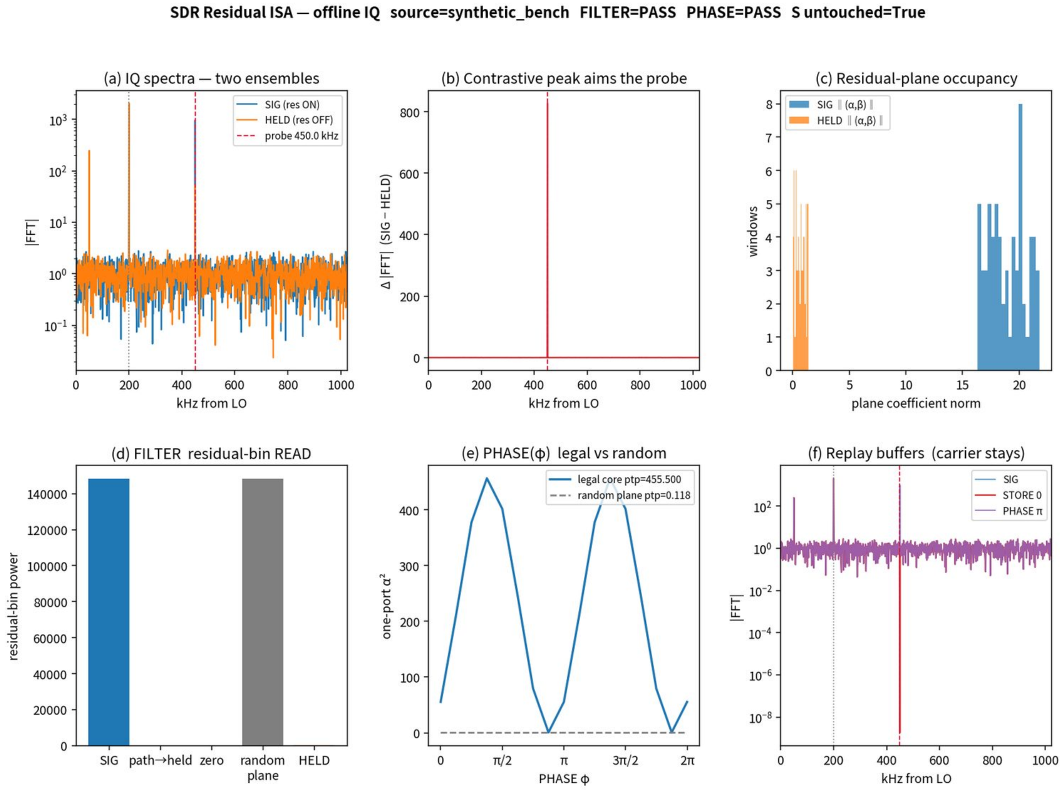

Figure 29. Offline SDR IQ residual drop-in. Hardware path is live; two-ensemble residual plant still needs a separate TX.

16. Superposition as Residual Stream

The classical mechanistic-interpretability claim is that features exist in superposition inside the residual stream: many features are packed into overlapping linear directions, and the residual stream is the place where they interfere and from which they are read out. The experiments above show that the same formal pattern appears in physical residual streams.

In every domain a residual component can be written as a low-dimensional direction (or subspace) that is approximately orthogonal to the dominant carrier, recovers the residual-dependent variance, and is causally necessary under path-patch. Sparse dictionary recovery isolates that component as an explicit atom whose coefficient can be set to zero or patched while the carrier remains intact. This is precisely the operational content of superposition: residual features share the same vector space with carriers, are not axis-aligned to them, yet can be selectively recovered and controlled.

What transferred: contrastive extraction; path-patch mediation; zero-collapse; random intact; selective residual-seeded atoms when the residual is concentrated; a pure-quadratic lift that surfaces second-moment residual structure linear ports miss; an original stream that can be left untouched while parallel ports work; an online write-hook that changes modular GPT-2 answer probability; PHASE/BEAM unitarity on a legal 2-D core; field and kernel ports after a linear continuum processor.

What did not transfer, or did not hold: Train A/B did not show that supervising the residual READ during training improves accuracy or FILTER rate under the written sidecar. Residual-seeded SAE on real Planck ports was selective but incomplete (36.76% retained). CRAI-2 mixture-domain suppress was not printable. CRAI-3 PASS is on a synthetic scene with a simple harmonic carrier. 96 kHz causal lock-in is legal on latency and weaker on isolation depth. QNN 14-D used reconstructed residual geometry, not a live dump. Closed-loop RF on owned hardware is defined and not yet run.

The claim is therefore not “everything is a transformer residual stream.” The claim is that an additive residual stream, once defined, admits the same addressing filter, and that processors on that stream are not themselves a new instruction set.

17. Limitations

Many early demonstrations are controlled synthetic mixtures with planted residuals. Real drop-ins exist (UAH Mueller, Zenodo NV ODMR, Planck R3.01, UCI naval, owned SDR IQ), but they are not yet a product. Statistical depth is multi-seed where the compute budget allowed (GPT-2 Phases A–D, Train A/B, QNN phase sweeps). Optical, NV, and CMB results are typically single-pipeline with internal random controls, not 16-seed hardware sweeps. The 14-D map is a fixed polynomial. Online write-hooks are demonstrated on modular GPT-2, not on natural-language models. Audio product quality is not a computer metric. Real-hardware quantum noise will degrade residual fidelity; dynamical decoupling and shadow tomography are natural next steps, not claims already made.

18. Status and Next Work

Closed. Optical classical suite including real Mueller. RF path-patch P1–P3. NV single / multi / B / T plus real Hole/NoHole 14-D. Planck residual ports. EEG and 5-D change coordinate. Telemetry synthetic + real naval. GPT-2 Phases A–D (16/16 online). Residual Stream ISA + interferometer. CRAI-3 synthetic PASS and 96 kHz causal bank. Continuum P1–P3. Train A/B executed and honestly failed the $B \geq A$ rule. Protocol v1.1 with Step 1b. First owned-SDR IQ dump.

In hand, not yet closed. RTL-SDR V4 receive-only plant. Real multi-track close-mics for CRAI. Vendor path (Sennheiser Connecticut and/or introduction to Shure) after an owned-plant proof.

Ordered next. (1) Owned-plant RF closed loop on the V4: conducted, RX-only SDR, separate TX, WRITE an actuator, S untouched. (2) Real unprocessed close-mics through CRAI before any plugin wrap. 96 kHz, one-in / one-out, zero-lookahead if isolation is printable. Plugin path only after the ear says yes: JUCE / VST3 / AU / CLAP Native. S6L AAX DSP / iLok / VENUE is a later Avid-dev step. (3) Same-architecture sidecar retune of Train A/B τ and lambdas only if that question is still live. (4) Further measured-field drop-ins when the file arrives.

Do not start a neural-operator training stack, a 1T physics model, or a third architecture inside A/B.

19. Conclusion

A residual causal toolkit — contrastive residual extraction, sparse dictionary / SAE recovery, selective ablation with matched random controls, path-patch mediation, and a small ISA on residual RAM — transfers from discrete transformer residual streams through narrowband RF, pure classical optical residual fields (synthetic and real Mueller), NV color-center residual streams, multi-channel telemetry, EEG, Planck CMB residuals, audio isolation, a continuum field processor, and mid-circuit quantum residual geometry. Residual components that live in superposition with carriers are recoverable, causally necessary, and selectively controllable. The residual stream is therefore not only the locus of superposition in neural networks; it is a general mixed-signal object whose residual components can be isolated and controlled by the same protocol.

From 17 August through 8 September 2026 the toolkit moved from a transformer-native idea to a domain-agnostic addressing filter with a small ISA, a field branch, a working offline audio isolator, an online residual write-hook that changes GPT-2 answer probability, and an owned SDR sitting on the bench. The method did not change. The addresses multiplied. The failed trainer test was kept in the record. The next proof that matters is not another synthetic PASS. It is a residual port on a wire that can be held, written to an actuator that can be controlled, with the received stream left alone.

Data Sources and Repositories

Public data. UAH Mueller-Matrix Spectropolarimeter (github.com/eejames2017/UAH-Mueller-Matrix-Spectropolarimeter). NV ODMR Zenodo 10.5281/zenodo.14697917 (Neu & Nimba). B-field ensemble Figshare 10.6084/m9.figshare.28788437. UCI Naval Propulsion Plants Condition Based Maintenance. Planck COM_PowerSpect_CMB R3.01 + theory R3.01.

Repositories. t2addonio/residual-causal-toolkit — this corpus. t2addonio/residual-stream-grokking — original grokking residual-component paper, left intact. t2addonio/isolate-rescue-grokking — isolate / rescue / freeze-at-transition.

Prior working documents folded into this paper. Mixed-Signal Residual Stream Methodology v1.1; Optical Residual Streams paper (17 Aug 2026); Superposition as Residual Stream (17 Aug 2026); Causal Phase and Magnitude Interventions on Quantum Residual Streams (Aug 2026); Residual Stream ISA one-pager (27 Aug 2026); Progress paper (4 Sep 2026); Phase D online residual control lock-in.

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UCI Machine Learning Repository. Condition Based Maintenance of Naval Propulsion Plants.

End of working paper. Figures regenerated from campaign artifacts in the working store. S was never overwritten.